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Photographing optical lens assembly, image capturing unit and electronic device

Abstract

A photographing optical lens assembly includes, in order from an object to an image side, a first lens group, a second lens group and a third lens group. The first lens group includes a first lens element and a second lens element. The second lens group includes a third lens element, a fourth lens element and a fifth lens element. The third lens group includes a sixth lens element, a seventh lens element and an eighth lens element. The first lens element has positive refractive power. The seventh lens element has an object-side surface and an image-side surface being both aspheric. The eighth lens element has an image-side surface being concave in a paraxial region thereof, wherein both an object-side surface and the image-side surface thereof are aspheric, and the image-side surface of the eighth lens element has at least one inflection point.

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Background/Summary

RELATED APPLICATIONS (1) This application is a divisional patent application of U.S. patent No. U.S. Pat. No. 12,072,469 B2, filed on Dec. 21, 2020, which is a continuation patent application of U.S. patent No. U.S. Pat. No. 10,908,392 B2, filed on Oct. 24, 2017, which is a continuation patent application of U.S. patent No. U.S. Pat. No. 9,835,822 B2, filed on Oct. 21, 2015, which claims priority to Taiwan patent No. 1553341, filed on Aug. 11, 2015, which is incorporated by reference herein in its entirety.

BACKGROUND

Technical Field

(1) The present disclosure relates to a photographing optical lens assembly, an image capturing unit and an electronic device, more particularly to a photographing optical lens assembly and an image capturing unit applicable to an electronic device.

Description of Related Art

(2) In recent years, with the popularity of electronic devices having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have reduced the pixel size of sensors, and compact optical systems have gradually evolved toward the field of

higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

(3) A conventional optical system employed in a portable electronic product mainly adopts a lens structure with fewer lens elements. Due to the popularity of mobile terminals with high-end specifications, such as smartphones, wearable devices and tablet personal computers, the requirements for high resolution and image quality increase significantly. With the popularity of the compact optical systems in the electronic devices, the demand of good image quality is also higher due to the advancement of the image sensor and software in the high-end electronic device. Thus, the conventional optical system does not meet the requirements of good image quality and compact size simultaneously.

SUMMARY

(4) According to one aspect of the present disclosure, a photographing optical lens assembly includes lens elements including, in order from an object side to an image side, a first lens group, a second lens group and a third lens group. The first lens group includes a first lens element and a second lens element. The second lens group includes a third lens element, a fourth lens element, and a fifth lens element. The third lens group includes a sixth lens element, a seventh lens element and an eighth lens element. The first lens element has positive refractive power. Both an object-side surface and an image-side surface of the seventh lens element are aspheric. The eighth lens element has an image-side surface being concave in a paraxial region thereof. Both an object-side surface and the image-side surface of the eighth lens element are aspheric. The image-side of the eighth lens element has at least one inflection point. There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other along an optical axis of the photographing optical lens assembly. When a focal length of the photographing optical lens assembly is f , a curvature radius of the image-side surface of the eighth lens element is $R16$, the following condition is satisfied:

$$0 < f/R16 < 6.5.$$

(5) According to another aspect of the present disclosure, an image capturing unit includes the aforementioned photographing optical lens assembly and an image sensor, wherein the image sensor is disposed on the image side of the photographing optical lens assembly.

(6) According to still another aspect of present disclosure, an electronic device includes the aforementioned image capturing unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure can be more understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

(2) FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure;

(3) FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment;

(4) FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of the present disclosure;

(5) FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment;

(6) FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure;

(7) FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment;

- (8) FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure;
- (9) FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment;
- (10) FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure;
- (11) FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment;
- (12) FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure;
- (13) FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment;
- (14) FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure;
- (15) FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 7th embodiment;
- (16) FIG. 15 is a schematic view of an image capturing unit according to the 8th embodiment of the present disclosure;
- (17) FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 8th embodiment;
- (18) FIG. 17 is a schematic view of an image capturing unit according to the 9th embodiment of the present disclosure;
- (19) FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 9th embodiment;
- (20) FIG. 19 is a schematic view of an image capturing unit according to the 10th embodiment of the present disclosure;
- (21) FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 10th embodiment;
- (22) FIG. 21 is a schematic view of an image capturing unit according to the 11th embodiment of the present disclosure;
- (23) FIG. 22 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 11th embodiment;
- (24) FIG. 23 is a schematic view of an image capturing unit according to the 12th embodiment of the present disclosure;
- (25) FIG. 24 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 12th embodiment;
- (26) FIG. 25 is a schematic view of BL, TL, Y11, Y82, Yc7 and Yc82 in FIG. 1;
- (27) FIG. 26 shows an electronic device according to one embodiment;
- (28) FIG. 27 shows an electronic device according to another embodiment; and
- (29) FIG. 28 shows an electronic device according to still another embodiment.

DETAILED DESCRIPTION

(30) A photographing optical lens assembly includes lens elements including, in order from an object side to an image side, a first lens group, a second lens group and a third lens group. The first lens group includes a first lens element and a second lens element. The second lens group includes a third lens element, a fourth lens element and a fifth lens element. The third lens group includes a sixth lens element, a seventh lens element and an eighth lens element.

(31) There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other; that is, each lens element of the third lens group is a single and non-cemented lens element. Moreover, the manufacturing process of the cemented lenses is more complex than the non-cemented lenses. In particular, an image-side surface of one lens

element and an object-side surface of the following lens element need to have accurate curvature to ensure these two lens elements will be highly cemented. However, during the cementing process, those two lens elements might not be highly cemented due to displacement and it is thereby not favorable for the image quality. Therefore, there is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other in the present disclosure for avoiding problems with cemented lens elements. In addition, the air gap in the paraxial region between every two lens elements of the first lens group, the second lens group and the third lens group that are adjacent to each other can be constant; that is, each lens elements of the photographing optical lens assembly in the paraxial region can be stationary relative to each other.

(32) The first lens group can have positive refractive power, the second lens group can have positive refractive power, and the third lens group can have negative refractive power. Therefore, it is favorable for applying the photographing optical lens assembly to the electronic devices having different sizes and different requirements of image resolution.

(33) The first lens element of the first lens group has positive refractive power. Therefore, it is favorable for providing the needed positive refractive power while reducing a total track length in the photographing optical lens assembly.

(34) The second lens element of the first lens group can have negative refractive power. The second lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for correcting the aberration of the first lens element so as to improve the image quality.

(35) The third, fourth and fifth lens elements of the second lens group and the sixth lens element of the third lens group can have positive or negative refractive power. Therefore, it is favorable for distributing the refractive power of the photographing optical lens assembly while correcting aberrations.

(36) The seventh lens element of the third lens group can have positive or negative refractive power. The seventh lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable with the principal point being positioned away from the image side of the photographing optical lens assembly for reducing a back focal length so as to maintain a compact size thereof.

(37) The eighth lens element of the third lens group can have negative refractive power. The eighth lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The image-side surface of the eighth lens element has at least one inflection point. Therefore, it is favorable for reducing the incident angle of the light on the image sensor so as to improve the image-sensing efficiency, and thereby correcting aberrations of the off-axis field.

(38) According to the present disclosure, the photographing optical lens assembly includes the first lens group, the second lens group and the third lens group. Therefore, the refractive power distribution of the first lens group, the second lens group and the third lens group is favorable for adjusting the design parameters of the photographing optical lens assembly. Furthermore, it is favorable for correcting aberrations. For example, when both the first lens element and the second lens element have positive refractive power with the first lens group having positive refractive power, it is favorable for the first and second lens elements having proper refractive power distribution while reducing the sensitivity of the photographing optical lens assembly. Also, when the first lens element has positive refractive power and the second lens element has negative refractive power, the first lens element and the second lens element with opposite refractive power are favorable for correcting aberrations.

(39) When a focal length of the photographing optical lens assembly is f , and a curvature radius of the image-side surface of the eighth lens element is $R16$, the following condition is satisfied: $0 < f/R16 < 6.5$. Therefore, it is favorable for reducing the back focal length of the photographing optical lens assembly so as to maintain a compact size thereof. Preferably, the following condition

can also be satisfied: $0.3 < f/R16 < 5.0$. More preferably, the following condition can also be satisfied: $1.0 < f/R16 < 4.5$.

(40) According to the present disclosure, the photographing optical lens assembly further includes a stop. When an axial distance between the stop and an image-side surface of the lens element closest to an image surface is SD, and an axial distance between an object-side surface of the lens element closest to an imaged object and the image-side surface of the lens element closest to the image surface is TD, the following condition can be satisfied: $0.70 < SD/TD < 1.10$. Therefore, it is favorable for properly placing the stop to provide sufficient field of view and obtain a balance between the total track length and the incident angle of the light on the image sensor.

(41) The photographing optical lens assembly may include one or more additional lens elements in any of the three lens groups to further enhance the image quality. That is, the photographing optical lens assembly may have more than eight lens elements.

(42) When a maximum refractive index among the lens elements of the photographing optical lens assembly is Nmax, the following condition can be satisfied: $1.55 < N_{max} < 1.70$. Therefore, it is favorable for designing the lens elements with more flexibility so as to correct the optical characteristics of the lens elements and improve the image quality.

(43) When an Abbe number of the first lens element is V1, an Abbe number of the second lens element is V2, and an Abbe number of the sixth lens element is V6, the following condition can be satisfied: $(V2+V6)/V1 < 1.0$. Therefore, the photographing optical lens assembly has balanced the capability for correcting chromatic aberration at both the object side and the image side.

(44) When an axial distance between the object-side surface of the lens element closest to the imaged object and the image surface is TL, and a maximum image height of the photographing optical lens assembly (half of a diagonal length of an effective photosensitive area of the image sensor) is ImgH, the following condition can be satisfied: $TL/ImgH < 2.1$. Therefore, it is favorable for simultaneously satisfying the requirement of compact size and a large image surface so as to apply the photographing optical lens assembly to electronic devices with high image resolution specifications. As seen in FIG. 25, FIG. 25 shows a schematic view of TL in FIG. 1. In FIG. 1, the first lens element is the lens element closest to the imaged object in this embodiment.

(45) When the axial distance between the object-side surface of the lens element closest to the imaged object and the image surface is TL, and an entrance pupil diameter of the photographing optical lens assembly is EPD, the following condition can be satisfied: $1.5 < TL/EPD < 4.0$.

Therefore, it is favorable for reducing the total track length of the photographing optical lens assembly and obtaining the characteristics of compactness and a large aperture.

(46) When half of a maximal field of view of the photographing optical lens assembly is HFOV, the following condition can be satisfied: $30.0 \text{ degrees} < HFOV < 50.0 \text{ degrees}$. Therefore, it is favorable for providing the photographing optical lens assembly with proper field of view.

(47) When the focal length of the photographing optical lens assembly is f, and a focal length of the first lens element is f1, the following condition can be satisfied: $0 < f/f1 < 2.5$. Therefore, the refractive power at the object side of the photographing optical lens assembly is sufficient for focusing the light rays, and thereby reducing the total track length. Preferably, the following condition can also be satisfied: $0 < f/f1 < 1.5$.

(48) When the axial distance between the object-side surface of the lens element closest to the imaged object and the image surface is TL, the following condition can be satisfied: $TL < 12.0 \text{ mm}$. Therefore, it is favorable for reducing the total track length so as to obtain compactness.

(49) When a curvature radius of the object-side surface of the eighth lens element is R15, and the curvature radius of the image-side surface of the eighth lens element is R16, the following condition can be satisfied: $-0.9 < (R15-R16)/(R15+R16) < 10$. Therefore, it is favorable for preventing the refractive power at the image side of the photographing optical lens assembly from overly strong and effectively correcting the astigmatism. Preferably, the following condition can also be satisfied: $-0.5 < (R15-R16)/(R15+R16) < 2.5$.

(50) When a maximum effective radius of the object-side surface of the first lens element is Y11, and a maximum effective radius of the image-side surface of the eighth lens element is Y82, the following condition can be satisfied: $0.20 < Y11/Y82 < 0.70$. Therefore, it is favorable for keep the photographing optical lens assembly compact and enlarging the field of view, and thereby satisfying the requirements of convenience and multifunction. As seen in FIG. 25, FIG. 25 shows a schematic view of Y11 and Y82 in FIG. 1.

(51) When a maximum axial distance among all axial distances between every two lens elements of the photographing optical lens assembly that are adjacent to each other is ATmax, and the maximum image height of the photographing optical lens assembly is ImgH, the following condition can be satisfied: $ATmax/ImgH < 0.30$. Therefore, it is favorable for arranging the size of each lens element so as to keep the photographing optical lens assembly compact.

(52) When the focal length of the photographing optical lens assembly is f, and the entrance pupil diameter of the photographing optical lens assembly is EPD, the following condition can be satisfied: $f/EPD < 2.60$. Therefore, it is favorable for providing sufficient amount of incident light so as to improve the image quality.

(53) When the focal length of the first lens element is f1, and a focal length of the second lens element is f2, the following condition can be satisfied: $-1.0 < f1/f2 < 0.7$. Therefore, the refractive power at the object side of the photographing optical lens assembly is sufficient for maintaining a compact size thereof.

(54) When an axial distance between the image-side surface of the lens element closest to the image surface and the image surface is BL, and the entrance pupil diameter of the photographing optical lens assembly is EPD, the following condition can be satisfied: $0.10 < BL/EPD < 0.70$. Therefore, it is favorable for controlling the back focal length and the entrance pupil to reduce the back focal length and to obtain the brightness simultaneously. As seen in FIG. 25, FIG. 25 shows a schematic view of BL in FIG. 1. In FIG. 1, the eighth lens element is the lens element closest to the image surface of the photographing optical lens assembly in this embodiment.

(55) When a sum of every axial distance between every two lens elements of the photographing optical lens assembly that are adjacent to each other is EAT, the axial distance between the image-side surface of the lens element closest to the image surface and the image surface is BL, and a sum of central thicknesses of the lens elements of the photographing optical lens assembly is ΣCT , the following condition can be satisfied: $(\Sigma AT + BL)/\Sigma CT < 0.80$. Therefore, it is favorable for tightly arranging the lens elements for compactness.

(56) When a curvature radius of an object-side surface of the sixth lens element is R11, and a curvature radius of an image-side surface of the sixth lens element is R12, the following condition can be satisfied: $-2.5 < (R11 - R12)/(R11 + R12) < 0.80$. Therefore, it is favorable for correcting the astigmatism and Petzval's Sum.

(57) When a vertical distance between a non-axial critical point on the object-side surface or the image-side surface of the seventh lens element and an optical axis is Yc7, and the focal length of the photographing optical lens assembly is f, the following condition can be satisfied: $0.10 < Yc7/f < 0.60$. Therefore, it is favorable for correcting the aberration of the off-axis field so as to improve the image quality at the off-axis region. Please refer to FIG. 25, FIG. 25 shows a schematic view of Yc7 in FIG. 1. A non-axial critical point is not located on the optical axis and its tangent is perpendicular to the optical axis. In FIG. 25, the non-axial critical point of the seventh lens element is located on the object-side surface, and there is no critical point on the image-side surface of the seventh lens element, but the present disclosure is not limited thereto. For example, both of the two surfaces of the seventh lens element can have critical point, and vertical distances between the critical points and the optical axis are both Yc7.

(58) When a vertical distance between a non-axial critical point on the image-side surface of the eighth lens element and the optical axis is Yc82, and the focal length of the photographing optical lens assembly is f, the following condition can be satisfied: $0.10 < Yc82/f < 0.80$. Therefore, it is

favorable for arranging the shape of the lens element at the image side so as to correct aberrations and increase relative illumination, and thereby improving the resolution at the off-axis region of the image. Please refer to FIG. 25, which is schematic view of Yc82 in FIG. 1.

(59) According to the present disclosure, the first lens element can have the strongest refractive power among the lens elements of the photographing optical lens assembly, that is, the absolute value of the refractive power of the first lens element is the greatest among the lens elements of the photographing optical lens assembly. Therefore, it is favorable for effectively reducing the back focal length of the photographing optical lens assembly.

(60) When the focal length of the photographing optical lens assembly is f , a focal length of the first lens group is $fG1$, a focal length of the second lens group is $fG2$, and a focal length of the third lens group is $fG3$, the following conditions can be satisfied: $0.1 < f/fG1 < 2.0$; $-0.4 < f/fG2 < 0.8$; and $-1.0 < f/fG3 < 1.0$. Therefore, the photographing optical lens assembly is favorably to be applied to different kinds of electronic devices by the refractive power distributions of the first lens group, the second lens group and the third lens group.

(61) When the focal length of the photographing optical lens assembly is f , and a focal length of the seventh lens element is $f7$, the following condition can be satisfied: $-1.0 < f/f7 < 2.8$. Therefore, it is favorable for providing sufficient refractive power at the image side of the photographing optical lens assembly so as to balance the arrangement of the lens elements of the photographing optical lens assembly, and thereby improving the image quality.

(62) When the focal length of the photographing optical lens assembly is f , and a focal length of the eighth lens element is $f8$, the following condition can be satisfied: $-2.8 < f/f8 < 1.0$. Therefore, it is favorable for correcting the distortion at the off-axis region of the image, and thereby improving the image quality.

(63) When a central thickness of the third lens element is $CT3$, and a central thickness of the fourth lens element is $CT4$, the following condition can be satisfied: $0.1 < CT3/CT4 < 4.0$. Therefore, it provides favorable moldability and homogeneity during the injection molding process.

(64) When a central thickness of the seventh lens element is $CT7$, and a central thickness of the eighth lens element is $CT8$, the following condition can be satisfied: $0.3 < CT7/CT8 < 4.0$. Therefore, it provides favorable moldability and homogeneity at the image side during the injection molding process.

(65) When an axial distance between the second lens element and the third lens element is $T23$, and an axial distance between the fifth lens element and the sixth lens element is $T56$, the following condition can be satisfied: $0.03 < T23/T56 < 6.00$. Therefore, it is favorable for assembling the lens elements with a higher manufacturing yield rate.

(66) When the axial distance between the second lens element and the third lens element is $T23$, and an axial distance between the seventh lens element and the eighth lens element is $T78$, the following condition can be satisfied: $0.03 < T23/T78 < 3.0$. Therefore, it is favorable for providing sufficient space between the adjacent lens elements so that the curvatures of the lens elements are more flexible to design, and thereby improving the capability for correcting aberrations at the off-axis field.

(67) According to the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can produce a telecentric effect by providing a longer distance between an exit pupil and the image surface and thereby improving the image-sensing efficiency of an image sensor (for example, CCD or CMOS). A middle stop disposed between the first lens element and the image surface is favorable for enlarging the view angle and thereby provides a wider field of view.

(68) According to the present disclosure, the lens elements of the photographing optical lens assembly can be made of glass or plastic material. When the lens elements are made of glass material, the refractive power distribution of the photographing optical lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can

be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the photographing optical lens assembly can also be reduced.

(69) According to the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface away from the paraxial region. Particularly, when the lens element has a convex surface, it indicates that the surface can be convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface can be concave in the paraxial region thereof. Moreover, when a region of refractive power or focus of a lens element is not defined, it indicates that the region of refractive power or focus of the lens element can be in the paraxial region thereof.

(70) According to the present disclosure, an image surface of the photographing optical lens assembly, based on the corresponding image sensor, can be flat or curved, especially a curved surface being concave facing towards the object side of the photographing optical lens assembly.

(71) According to the present disclosure, the photographing optical lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is allocated for eliminating the stray light and thereby improving the image quality thereof.

(72) According to the present disclosure, an image capturing unit is provided. The image capturing unit includes the photographing optical lens assembly according to the aforementioned photographing optical lens assembly of the present disclosure, and an image sensor, wherein the image sensor is disposed on the image side of the aforementioned photographing optical lens assembly, that is, the image sensor can be disposed on or near an image surface of the aforementioned photographing optical lens assembly. In some embodiments, the image capturing unit can further include a barrel member, a holding member or a combination thereof.

(73) In FIG. 26, FIG. 27, and FIG. 28, an image capturing unit **10** may be installed in, but not limited to, an electronic device, including a smart phone (FIG. 26), a tablet personal computer (FIG. 27) or a wearable device (FIG. 28). The electronic devices shown in the figures are only exemplary for showing the image capturing unit of the present disclosure installed in an electronic device and are not limited thereto. In some embodiments, the electronic device can further include, but not limited to, a display unit, a control unit, a storage unit, a random access memory unit (RAM), a read only memory unit (ROM) or a combination thereof.

(74) According to the present disclosure, the photographing optical lens assembly can be optionally applied to optical systems with a movable focus. Furthermore, the photographing optical lens assembly is featured with good capability in the aberration correction and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices, smart televisions, network surveillance devices, motion sensing input devices, dashboard cameras, vehicle backup cameras and other electronic imaging devices. According to the above description of the present disclosure, the following specific embodiments are provided for further explanation.

1st Embodiment

(75) FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment. In FIG. 1, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor **197**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **100**, a first lens element **110**, a second lens element **120**, a third lens element **130**, a fourth lens element **140**, a

fifth lens element **150**, a sixth lens element **160**, a seventh lens element **170**, an eighth lens element **180**, an IR-cut filter **190** and an image surface **195**. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element **110** and the second lens element **120**. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element **130**, fourth lens element **140** and the fifth lens element **150**. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element **160**, the seventh lens element **170** and the eighth lens element **180**. The image sensor **197** is disposed on or near the image surface **195** of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (**110-180**). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(76) The first lens element **110** with positive refractive power has an object-side surface **111** being convex in a paraxial region thereof and an image-side surface **112** being concave in a paraxial region thereof. The first lens element **110** is made of plastic material and has the object-side surface **111** and the image-side surface **112** being both aspheric.

(77) The second lens element **120** with negative refractive power has an object-side surface **121** being convex in a paraxial region thereof and an image-side surface **122** being concave in a paraxial region thereof. The second lens element **120** is made of plastic material and has the object-side surface **121** and the image-side surface **122** being both aspheric.

(78) The third lens element **130** with positive refractive power has an object-side surface **131** being convex in a paraxial region thereof and an image-side surface **132** being convex in a paraxial region thereof. The third lens element **130** is made of plastic material and has the object-side surface **131** and the image-side surface **132** being both aspheric.

(79) The fourth lens element **140** with negative refractive power has an object-side surface **141** being concave in a paraxial region thereof and an image-side surface **142** being convex in a paraxial region thereof. The fourth lens element **140** is made of plastic material and has the object-side surface **141** and the image-side surface **142** being both aspheric.

(80) The fifth lens element **150** with negative refractive power has an object-side surface **151** being concave in a paraxial region thereof and an image-side surface **152** being convex in a paraxial region thereof. The fifth lens element **150** is made of plastic material and has the object-side surface **151** and the image-side surface **152** being both aspheric.

(81) The sixth lens element **160** with negative refractive power has an object-side surface **161** being concave in a paraxial region thereof and an image-side surface **162** being convex in a paraxial region thereof. The sixth lens element **160** is made of plastic material and has the object-side surface **161** and the image-side surface **162** being both aspheric.

(82) The seventh lens element **170** with positive refractive power has an object-side surface **171** being convex in a paraxial region thereof and an image-side surface **172** being convex in a paraxial region thereof. The seventh lens element **170** is made of plastic material and has the object-side surface **171** and the image-side surface **172** being both aspheric.

(83) The eighth lens element **180** with negative refractive power has an object-side surface **181** being convex in a paraxial region thereof and an image-side surface **182** being concave in a paraxial region thereof. The eighth lens element **180** is made of plastic material and has the object-side surface **181** and the image-side surface **182** being both aspheric. The image-side surface **182** has at least one inflection point.

(84) The IR-cut filter **190** is made of glass and located between the eighth lens element **180** and the image surface **195**, and will not affect the focal length of the photographing optical lens assembly.

(85) The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

- (86) $X(Y) = (Y^2 / R) / (1 + \sqrt{1 - (1 + k) \times (Y / R)^2}) + \sum_i A_i \times (Y^i)$, where, X is the relative distance between a point on the aspheric surface spaced at a distance Y from an optical axis and the tangential plane at the aspheric surface vertex on the optical axis; Y is the vertical distance from the point on the aspheric surface to the optical axis; R is the curvature radius; k is the conic coefficient; and A_i is the i-th aspheric coefficient, and in the embodiments, i may be, but is not limited to, 4, 6, 8, 10, 12, 14 and 16.
- (87) In the photographing optical lens assembly of the image capturing unit according to the 1st embodiment, when a focal length of the photographing optical lens assembly is f, an f-number of the photographing optical lens assembly is Fno, and half of a maximal field of view of the photographing optical lens assembly is HFOV, these parameters have the following values: f=3.75 millimeters (mm); Fno=1.90; and HFOV=36.8 degrees (deg.).
- (88) When a maximum refractive index among the lens elements of the photographing optical lens assembly is Nmax, the following condition is satisfied: Nmax=1.639. In this embodiment, Nmax is a maximum refractive index among the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, the sixth lens element **160**, the seventh lens element **170** and the eighth lens element **180**.
- (89) When an Abbe number of the first lens element **110** is V1, an Abbe number of the second lens element **120** is V2, an Abbe number of the sixth lens element is V6, the following condition is satisfied: $(V2+V6)/V1=0.84$.
- (90) When a curvature radius of the object-side surface **161** of the sixth lens element **160** is R11, a curvature radius of the image-side surface **162** of the sixth lens element **160** is R12, the following condition is satisfied: $(R11-R12)/(R11+R12)=-0.13$.
- (91) When a curvature radius of the object-side surface **181** of the eighth lens element **180** is R15, a curvature radius of the image-side surface **182** of the eighth lens element **180** is R16, the following condition is satisfied: $(R15-R16)/(R15+R16)=-0.77$.
- (92) When the focal length of the photographing optical lens assembly is f, the curvature radius of the image-side surface **182** of the eighth lens element **180** is R16, the following condition is satisfied: $f/R16=2.61$.
- (93) When the focal length of the photographing optical lens assembly is f, a focal length of the first lens element **110** is f1, the following condition is satisfied: $f/f1=1.11$.
- (94) When the focal length of the first lens element **110** is f1, a focal length of the second lens element **120** is f2, the following condition is satisfied: $f1/f2=-0.56$.
- (95) When a maximum effective radius of the object-side surface **111** of the first lens element **110** is Y11, a maximum effective radius of the image-side surface **182** of the eighth lens element **180** is Y82, the following condition is satisfied: $Y11/Y82=0.43$.
- (96) When an axial distance between the stop **100** and an image-side surface of the lens element closest to the image surface **195** is SD, an axial distance between an object-side surface of the lens element closest to an imaged object and the image-side surface of the lens element closest to the image surface **195** is TD, the following condition is satisfied: $SD/TD=0.92$. In this embodiment, SD is an axial distance between the stop **100** and the image side-surface **182** of the eighth lens element **180**, and TD is an axial distance between the object-side surface **111** of the first lens element **110** and the image side-surface **182** of the eighth lens element **180**.
- (97) When a maximum axial distance among all axial distances between every two lens elements of the photographing optical lens assembly that are adjacent to each other is ATmax, a maximum image height of the photographing optical lens assembly is ImgH, the following condition is satisfied: $ATmax/ImgH=0.11$. In this embodiment, ATmax is a maximum axial distance among axial distances between every two first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, the sixth lens element **160**, the seventh lens element **170** and the eighth lens element **180** that are adjacent to each other.

(98) When the focal length of the photographing optical lens assembly is f , an entrance pupil diameter of the photographing optical lens assembly is EPD, the following condition is satisfied: $f/EPD=1.90$.

(99) When an axial distance between the image-side surface of the lens element closest to the image surface **195** and the image surface **195** is BL, the entrance pupil diameter of the photographing optical lens assembly is EPD, the following condition is satisfied: $BL/EPD=0.39$. In this embodiment, BL is an axial distance between the image-side surface **182** of the eighth lens element **180** and the image surface **195**.

(100) When an axial distance between the object-side surface of the lens element closest to the imaged object and the image surface **195** is TL, the entrance pupil diameter of the photographing optical lens assembly is EPD, the following condition is satisfied: $TL/EPD=2.42$. In this embodiment, TL is an axial distance between the object-side surface **111** of the first lens element **110** and the image surface **195**.

(101) When a sum of every axial distance between every two lens elements of the photographing optical lens assembly that are adjacent to each other is ΣAT , the axial distance between the image-side surface of the lens element closest to the image surface **195** and the image surface **195** is BL, a sum of central thicknesses of lens elements of the photographing optical lens assembly is ΣCT , the following condition is satisfied: $(\Sigma AT+BL)/\Sigma CT=0.59$. In this embodiment, ΣAT is a sum of every axial distance between every two of the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, the sixth lens element **160**, the seventh lens element **170** and the eighth lens element **180** that are adjacent to each other, and ΣCT is a sum of central thicknesses of the first lens element **110** through the eighth lens element **180**.

(102) When the axial distance between the object-side surface of the lens element closest to the imaged object and the image surface **195** is TL, the maximum image height of the photographing optical lens assembly is $ImgH$, the following condition is satisfied: $TL/ImgH=1.65$.

(103) When the axial distance between the object-side surface of the lens element closest to the imaged object and the image surface **195** is TL, the following condition is satisfied: $TL=4.78$ mm.

(104) When a vertical distance between a non-axial critical point on the object-side surface **171** or the image-side surface **172** of the seventh lens element **170** and an optical axis is $Yc7$, the focal length of the photographing optical lens assembly is f , the following condition is satisfied: $Yc7/f=0.23$. In this embodiment, the non-axial critical point is located on the object-side surface **171** of the seventh lens element **170**, and there is no non-axial critical point on the image-side surface **172** of the seventh lens element **170**.

(105) When a vertical distance between a non-axial critical point on the image-side surface **182** of the eighth lens element **180** and the optical axis is $Yc82$, the focal length of the photographing optical lens assembly is f , the following condition is satisfied: $Yc82/f=0.29$.

(106) When the focal length of the photographing optical lens assembly is f , a focal length of the first lens group is $fG1$, the following condition is satisfied: $f/fG1=0.65$.

(107) When the focal length of the photographing optical lens assembly is f , a focal length of the second lens group is $fG2$, the following condition is satisfied: $f/fG2=0.45$.

(108) When the focal length of the photographing optical lens assembly is f , a focal length of the third lens group is $fG3$, the following condition is satisfied: $f/fG3=-0.41$.

(109) When the focal length of the photographing optical lens assembly is f , a focal length of the seventh lens element **170** is $f7$, the following condition is satisfied: $f/f7=0.86$.

(110) When the focal length of the photographing optical lens assembly is f , a focal length of the eighth lens element **180** is $f8$, the following condition is satisfied: $f/f8=-1.20$.

(111) When a central thickness of the third lens element **130** is $CT3$, a central thickness of the fourth lens element **140** is CT , the following condition is satisfied: $CT3/CT4=1.60$.

(112) When a central thickness of the seventh lens element **170** is $CT7$, a central thickness of the

eighth lens element **180** is CT8, the following condition is satisfied: $CT7/CT8=2.39$.

(113) When an axial distance between the second lens element **120** and the third lens element **130** is T23, an axial distance between the fifth lens element **150** and the sixth lens element **160** is T56, the following condition is satisfied: $T23/T56=1.46$.

(114) When the axial distance between the second lens element **120** and the third lens element **130** is T23, an axial distance between the seventh lens element **170** and the eighth lens element **180** is T78, the following condition is satisfied: $T23/T78=0.80$.

(115) The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

(116) TABLE-US-00001 TABLE 1 1st Embodiment $f = 3.75$ mm, $Fno = 1.90$, $HFOV = 36.8$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.325 2 Lens 1 1.685 (ASP) 0.546 Plastic 1.544 55.9 3.37 3 18.567 (ASP)
0.070 4 Lens 2 2.923 (ASP) 0.264 Plastic 1.639 23.5 -6.03 5 1.604 (ASP) 0.253 6 Lens 3 6.358
(ASP) 0.426 Plastic 1.544 55.9 7.75 7 -12.215 (ASP) 0.050 8 Lens 4 -10.576 (ASP) 0.266 Plastic
1.544 55.9 -430.85 9 -11.174 (ASP) 0.104 10 Lens 5 -10.367 (ASP) 0.251 Plastic 1.544 55.9
-274.67 11 -11.235 (ASP) 0.173 12 Lens 6 -2.060 (ASP) 0.220 Plastic 1.639 23.5 -16.13 13
-2.681 (ASP) 0.035 14 Lens 7 3.461 (ASP) 0.732 Plastic 1.544 55.9 4.37 15 -7.055 (ASP) 0.317
16 Lens 8 11.111 (ASP) 0.306 Plastic 1.535 55.7 -3.12 17 1.438 (ASP) 0.350 18 IR-cut filter Plano
0.210 Glass 1.517 64.2 — 19 Plano 0.206 20 Image Plano — Note: Reference wavelength is 587.6
nm (d-line).

(117) TABLE-US-00002 TABLE 2 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k = 1.7681E-01$
-1.2776E+01 -2.9857E+01 -1.2709E+01 -2.7741E+01 9.0000E+01 $A4 = 6.5549E-03$
-1.3129E-01 -1.6384E-01 1.3995E-01 -9.3270E-03 5.8849E-04 $A6 = -2.1368E-02$
3.6675E-01 4.0304E-01 -2.2455E-01 -4.4137E-03 -1.7985E-02 $A8 = 5.4198E-02$
-4.9085E-01 -5.1213E-01 4.7697E-01 2.5916E-02 -4.2249E-03 $A10 = -6.4445E-02$
3.7784E-01 3.7035E-01 -5.4385E-01 -4.1583E-02 -8.5112E-03 $A12 = 4.4523E-02$
-1.5437E-01 -1.5562E-01 3.2179E-01 2.6550E-02 — $A14 = -1.1704E-02$ 2.4066E-02
2.6062E-02 -6.9334E-02 -2.2918E-03 — Surface # 8 9 10 11 12 13 $k = 6.9735E+01$
8.3732E+01 6.2105E+01 7.4238E+01 1.0353E+00 -1.0145E+01 $A4 = -2.0966E-02$
-3.0502E-02 -5.0212E-02 -1.4195E-01 -4.4139E-02 -1.6893E-01 $A6 = 1.6769E-03$
-2.9339E-02 -4.1045E-02 1.1718E-01 2.4240E-01 2.6139E-01 $A8 = -7.7153E-03$
-3.2530E-03 -2.1920E-02 -1.0703E-01 -4.2265E-01 -3.0327E-01 $A10 = -1.0660E-03$
-6.6123E-03 4.7669E-03 -1.1645E-01 4.0198E-01 2.1787E-01 $A12 = 1.4550E-04$
1.0091E-06 3.1028E-01 -1.7230E-01 -8.0449E-02 $A14 = — — — -2.0573E-01$
2.4455E-02 1.1717E-02 $A16 = — — — 4.5154E-02 — —$ Surface # 14 15 16 17 $k =$
-4.8924E+01 -7.9584E+01 1.4669E+01 -3.1378E+00 $A4 = -3.0918E-02$ 4.9387E-03
-3.4608E-01 -2.4817E-01 $A6 = 3.7542E-02$ 1.2692E-02 1.8930E-01 1.6312E-01 $A8 =$
-8.5743E-02 -2.6623E-02 -5.8697E-02 -7.5256E-02 $A10 = 5.2991E-02$ 1.1128E-02
1.3504E-02 2.2753E-02 $A12 = -1.7634E-02$ -1.7350E-03 -2.2740E-03 -4.1789E-03 $A14 =$
2.5249E-03 8.2882E-05 2.3106E-04 4.1494E-04 $A16 = — — -1.0250E-05$ -1.6950E-05

(118) In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-20 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. $A4$ - $A16$ represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

(119) FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of

the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment. In FIG. 3, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor **297**. The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element **210**, an aperture stop **200**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a fifth lens element **250**, a sixth lens element **260**, a seventh lens element **270**, an eighth lens element **280**, an IR-cut filter **290** and an image surface **295**. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element **210** and the second lens element **220**. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element **230**, fourth lens element **240** and the fifth lens element **250**. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element **260**, the seventh lens element **270** and the eighth lens element **280**. The image sensor **297** is disposed on or near the image surface **295** of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (**210-280**). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(120) The first lens element **210** with positive refractive power has an object-side surface **211** being convex in a paraxial region thereof and an image-side surface **212** being concave in a paraxial region thereof. The first lens element **210** is made of plastic material and has the object-side surface **211** and the image-side surface **212** being both aspheric.

(121) The second lens element **220** with negative refractive power has an object-side surface **221** being convex in a paraxial region thereof and an image-side surface **222** being concave in a paraxial region thereof. The second lens element **220** is made of plastic material and has the object-side surface **221** and the image-side surface **222** being both aspheric.

(122) The third lens element **230** with positive refractive power has an object-side surface **231** being convex in a paraxial region thereof and an image-side surface **232** being convex in a paraxial region thereof. The third lens element **230** is made of plastic material and has the object-side surface **231** and the image-side surface **232** being both aspheric.

(123) The fourth lens element **240** with negative refractive power has an object-side surface **241** being concave in a paraxial region thereof and an image-side surface **242** being convex in a paraxial region thereof. The fourth lens element **240** is made of plastic material and has the object-side surface **241** and the image-side surface **242** being both aspheric.

(124) The fifth lens element **250** with positive refractive power has an object-side surface **251** being concave in a paraxial region thereof and an image-side surface **252** being convex in a paraxial region thereof. The fifth lens element **250** is made of plastic material and has the object-side surface **251** and the image-side surface **252** being both aspheric.

(125) The sixth lens element **260** with negative refractive power has an object-side surface **261** being concave in a paraxial region thereof and an image-side surface **262** being convex in a paraxial region thereof. The sixth lens element **260** is made of plastic material and has the object-side surface **261** and the image-side surface **262** being both aspheric.

(126) The seventh lens element **270** with positive refractive power has an object-side surface **271** being convex in a paraxial region thereof and an image-side surface **272** being convex in a paraxial region thereof. The fourth lens element **270** is made of plastic material and has the object-side surface **271** and the image-side surface **272** being both aspheric.

(127) The eighth lens element **280** with negative refractive power has an object-side surface **281** being convex in a paraxial region thereof and an image-side surface **282** being concave in a paraxial region thereof. The eighth lens element **280** is made of plastic material and has the object-

side surface **281** and the image-side surface **282** being both aspheric. The image-side surface **282** of the eighth lens element **280** has at least one inflection point

(128) The IR-cut filter **290** is made of glass and located between the eighth lens element **280** and the image surface **295**, and will not affect the focal length of the photographing optical lens assembly.

(129) The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

(130) TABLE-US-00003 TABLE 3 2nd Embodiment $f = 3.77$ mm, $Fno = 1.95$, $HFOV = 36.7$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Lens 1 1.696 (ASP) 0.670 Plastic 1.544 56.0 3.41 2 16.917 (ASP) 0.046 3 Ape. Stop Plano 0.021
4 Lens 2 2.870 (ASP) 0.264 Plastic 1.660 20.4 -6.65 5 1.671 (ASP) 0.239 6 Lens 3 7.476 (ASP)
0.398 Plastic 1.544 56.0 8.39 7 -11.489 (ASP) 0.059 8 Lens 4 -10.025 (ASP) 0.234 Plastic 1.544
56.0 -56.87 9 -14.952 (ASP) 0.110 10 Lens 5 -13.260 (ASP) 0.223 Plastic 1.544 56.0 120.26 11
-11.091 (ASP) 0.158 12 Lens 6 -2.092 (ASP) 0.253 Plastic 1.660 20.4 -45.32 13 -2.358 (ASP)
0.035 14 Lens 7 4.446 (ASP) 0.678 Plastic 1.535 55.7 5.11 15 -6.733 (ASP) 0.316 16 Lend 8
11.265 (ASP) 0.322 Plastic 1.535 55.7 -3.00 17 1.391 (ASP) 0.350 18 IR-cut filter Plano 0.210
Glass 1.517 64.2 — 19 Plano 0.190 20 Image Plano — Note: Reference wavelength is 587.6 nm
(d-line).

(131) TABLE-US-00004 TABLE 4 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k = 1.4673E-01$
-1.5015E+01 -2.6270E+01 -1.3602E+01 -4.0052E+01 9.0000E+01 $A4 = 5.0324E-03$
-1.3146E-01 -1.5925E-01 1.3340E-01 -1.2275E-02 4.4223E-03 $A6 = -2.1430E-02$
3.6545E-01 4.0232E-01 -2.2625E-01 -7.4975E-03 -2.1088E-02 $A8 = 5.2679E-02$
-4.9019E-01 -5.1523E-01 4.7697E-01 2.4265E-02 -4.9957E-03 $A10 = -6.5303E-02$
3.7771E-01 3.7522E-01 -5.4666E-01 -3.8760E-02 -1.3338E-02 $A12 = 4.4160E-02$
-1.5424E-01 -1.5538E-01 3.2165E-01 2.6371E-02 — $A14 = -1.1692E-02$ 2.4061E-02
2.6069E-02 -6.9342E-02 -2.3186E-03 — Surface # 8 9 10 11 12 13 $k = 7.1376E+01$
8.2162E+01 6.1940E+01 7.3976E+01 9.4068E-01 -1.2659E+01 $A4 = -2.5943E-02$
-3.5396E-02 -5.7194E-02 -1.4323E-01 -4.4477E-02 -1.7166E-01 $A6 = 1.3875E-03$
-3.0765E-02 -3.9764E-02 1.1687E-01 2.4144E-01 2.5838E-01 $A8 = -1.0133E-02$
1.3015E-03 -2.2952E-02 -1.0712E-01 -4.2349E-01 -3.0376E-01 $A10 = -6.8767E-04$
-4.4100E-03 4.2506E-03 -1.1602E-01 4.0156E-01 2.1792E-01 $A12 = —$ 1.8038E-04
-5.3802E-05 3.1081E-01 -1.7210E-01 -8.0391E-02 $A14 = — — —$ -2.0572E-01
2.4497E-02 1.1738E-02 $A16 = — — —$ 4.5165E-02 — — Surface # 14 15 16 17 $k =$
-4.5933E+01 -8.3161E+01 1.3558E+01 -3.2789E+00 $A4 = -2.9810E-02$ 4.6801E-03
-3.4564E-01 -2.4764E-01 $A6 = 3.9769E-02$ 1.2493E-02 1.8932E-01 1.6278E-01 $A8 =$
-8.6250E-02 -2.6607E-02 -5.8752E-02 -7.5303E-02 $A10 = 5.3110E-02$ 1.1105E-02
1.3508E-02 2.2754E-02 $A12 = -1.7548E-02$ -1.7322E-03 -2.2726E-03 -4.1792E-03 $A14 =$
2.4697E-03 8.5269E-05 2.3129E-04 4.1491E-04 $A16 = — —$ -1.0258E-05 -1.6943E-05

(132) In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

(133) Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

(134) TABLE-US-00005 2nd Embodiment f [mm] 3.77 ($\Sigma AT + BL$)/ ΣCT 0.57 Fno 1.95 $TL/ImgH$ 1.65 $HFOV$ [deg.] 36.7 TL [mm] 4.78 $Nmax$ 1.660 $Yc7/f$ (object-side surface 0.23 271) ($V2 + V6$)/ $V1$ 0.73 $Yc7/f$ (image-side surface — 272) ($R11 - R12$)/($R11 + R12$) -0.06 $Yc82/f$ 0.29 ($R15 - R16$)/($R15 + R16$) 0.78 $f/fG1$ 0.70 $f/R16$ 2.71 $f/fG2$ 0.41 $f/f1$ 1.10 $f/fG3$ -0.44 $f1/f2$ -0.51 $f/f7$ 0.74 $Y11/Y82$ 0.47 $f/f8$ -1.26 SD/TD 0.82 $CT3/CT4$ 1.70 $ATmax/ImgH$ 0.11 $CT7/CT8$ 2.11 f/EPD 1.95

3rd Embodiment

(135) FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment. In FIG. 5, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor **397**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **300**, a first lens element **310**, a second lens element **320**, a third lens element **330**, a fourth lens element **340**, a fifth lens element **350**, a sixth lens element **360**, a seventh lens element **370**, an eighth lens element **380**, an IR-cut filter **390** and an image surface **395**. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element **310** and the second lens element **320**. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element **330**, fourth lens element **340** and the fifth lens element **350**. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element **360**, the seventh lens element **370** and the eighth lens element **380**. The image sensor **397** is disposed on or near the image surface **395** of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (**310-380**). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(136) The first lens element **310** with positive refractive power has an object-side surface **311** being convex in a paraxial region thereof and an image-side surface **312** being concave in a paraxial region thereof. The first lens element **310** is made of plastic material and has the object-side surface **311** and the image-side surface **312** being both aspheric.

(137) The second lens element **320** with negative refractive power has an object-side surface **321** being convex in a paraxial region thereof and an image-side surface **322** being concave in a paraxial region thereof. The second lens element **320** is made of plastic material and has the object-side surface **321** and the image-side surface **322** being both aspheric.

(138) The third lens element **330** with positive refractive power has an object-side surface **331** being convex in a paraxial region thereof and an image-side surface **332** being convex in a paraxial region thereof. The third lens element **330** is made of plastic material and has the object-side surface **331** and the image-side surface **332** being both aspheric.

(139) The fourth lens element **340** with negative refractive power has an object-side surface **341** being concave in a paraxial region thereof and an image-side surface **342** being concave in a paraxial region thereof. The fourth lens element **340** is made of plastic material and has the object-side surface **341** and the image-side surface **342** being both aspheric.

(140) The fifth lens element **350** with positive refractive power has an object-side surface **351** being convex in a paraxial region thereof and an image-side surface **352** being convex in a paraxial region thereof. The fifth lens element **350** is made of plastic material and has the object-side surface **351** and the image-side surface **352** being both aspheric.

(141) The sixth lens element **360** with negative refractive power has an object-side surface **361** being concave in a paraxial region thereof and an image-side surface **362** being convex in a paraxial region thereof. The sixth lens element **360** is made of plastic material and has the object-side surface **361** and the image-side surface **362** being both aspheric.

(142) The seventh lens element **370** with negative refractive power has an object-side surface **371** being convex in a paraxial region thereof and an image-side surface **372** being concave in a paraxial region thereof. The seventh lens element **370** is made of plastic material and has the object-side surface **371** and the image-side surface **372** being both aspheric.

(143) The eighth lens element **380** with positive refractive power has an object-side surface **381** being convex in a paraxial region thereof and an image-side surface **382** being concave in a paraxial region thereof. The eighth lens element **380** is made of plastic material and has the object-side surface **381** and the image-side surface **382** being both aspheric. The image-side surface **382** of the eighth lens element **380** has at least one inflection point.

(144) The IR-cut filter **390** is made of glass and located between the eighth lens element **380** and the image surface **395**, and will not affect the focal length of the photographing optical lens assembly.

(145) The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

(146) TABLE-US-00006 TABLE 5 3rd Embodiment $f = 3.97$ mm, $Fno = 1.75$, HFOV = 35.2 deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.415 2 Lens 1 1.766 (ASP) 0.650 Plastic 1.544 55.9 3.45 3 25.751 (ASP)
0.082 4 Lens 2 2.600 (ASP) 0.220 Plastic 1.639 23.5 -5.97 5 1.495 (ASP) 0.297 6 Lens 3 5.377
(ASP) 0.366 Plastic 1.544 55.9 7.25 7 -14.478 (ASP) 0.037 8 Lens 4 -11.768 (ASP) 0.147 Plastic
1.544 55.9 -8.92 9 8.291 (ASP) 0.087 10 Lens 5 9.624 (ASP) 0.354 Plastic 1.544 55.9 9.50 11
-11.034 (ASP) 0.236 12 Lens 6 -2.044 (ASP) 0.268 Plastic 1.639 23.5 -9.71 13 -3.204 (ASP)
0.035 14 Lens 7 2.047 (ASP) 0.419 Plastic 1.544 55.9 -122.49 15 1.843 (ASP) 0.243 16 Lens 8
1.346 (ASP) 0.587 Plastic 1.544 55.9 26.64 17 1.256 (ASP) 0.500 18 IR-cut filter Plano 0.210
Glass 1.517 64.2 — 19 Plano 0.243 20 Image Plano — Note: Reference wavelength is 587.6 nm
(d-line).

(147) TABLE-US-00007 TABLE 6 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k = 1.7606E-01$
-5.0000E+01 -3.5839E+01 -1.3306E+01 -2.4164E+01 9.0000E+01 $A4 = 3.3883E-03$
-1.3351E-01 -1.7221E-01 1.1742E-01 -1.5095E-02 -1.6332E-02 $A6 = -2.1413E-02$
3.6480E-01 4.0332E-01 -2.3781E-01 -1.2905E-02 -3.5827E-04 $A8 = 5.2368E-02$
-4.9306E-01 -5.1231E-01 4.8887E-01 2.8967E-02 — $A10 = -6.5934E-02$ 3.7906E-01
3.8118E-01 -5.5380E-01 -5.0574E-02 — $A12 = 4.3209E-02$ -1.5437E-01 -1.5562E-01
3.2179E-01 2.6550E-02 — $A14 = -1.1704E-02$ 2.4065E-02 2.6062E-02 -6.9334E-02
-2.2918E-03 — Surface # 8 9 10 11 12 13 $k = 2.7907E+01$ 3.1895E+01 -9.0000E+01
6.9188E+01 1.2718E+00 -9.9934E+00 $A4 = -2.6896E-03$ -1.1708E-02 -3.1201E-02
-1.1150E-01 -3.6883E-02 -1.8828E-01 $A6 = -7.7490E-03$ -2.6165E-02 -2.5688E-02
1.1406E-01 2.4728E-01 2.6313E-01 $A8 = -6.2458E-03$ -1.9250E-02 -1.0573E-01
-4.2313E-01 -3.0195E-01 $A10 = -8.8311E-03$ 8.8072E-03 -1.1698E-01 4.0336E-01
2.1802E-01 $A12 = -1.4562E-04$ 9.6476E-07 3.0921E-01 -1.7202E-01 -8.0454E-02 $A14 = -2.0573E-01$
2.4455E-02 1.1693E-02 $A16 = 4.5154E-02$ — — Surface #
14 15 16 17 $k = -1.0051E+01$ -1.4423E+01 -2.7322E+00 -1.7788E+00 $A4 = -4.1773E-02$
-1.8604E-02 -3.8002E-01 -2.8376E-01 $A6 = 3.1988E-02$ 1.2534E-02 1.9090E-01
1.6714E-01 $A8 = -8.2642E-02$ -2.6769E-02 -5.8388E-02 -7.5232E-02 $A10 = 5.1053E-02$
1.1185E-02 1.3534E-02 2.2738E-02 $A12 = -1.8106E-02$ -1.7156E-03 -2.2724E-03
-4.1790E-03 $A14 = 2.9196E-03$ 8.4354E-05 2.3092E-04 4.1504E-04 $A16 = -1.0325E-05$
-1.6969E-05

(148) In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

(149) Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

(150) TABLE-US-00008 3rd Embodiment f [mm] 3.97 ($\Sigma AT + BL$)/ ΣCT 0.65 Fno 1.75 $TL/ImgH$ 1.72 HFOV [deg.] 35.2 TL [mm] 4.98 $Nmax$ 1.639 $Yc7/f(\text{object-side surface } 0.24 \text{ } 371)$ ($V2 +$

V6)/V1 0.84 Yc7/f (image-side surface 0.29 372) (R11 - R12)/(R11 + R12) -0.22 Yc82/f 0.29 (R15 - R16)/(R15 + R16) 0.03 f/fG1 0.65 f/R16 3.16 f/fG2 0.52 f/f1 1.15 f/fG3 -0.40 f1/f2 -0.58 f/f7 -0.03 Y11/Y82 0.48 f/f8 0.15 SD/TD 0.90 CT3/CT4 2.49 ATmax/ImgH 0.10 CT7/CT8 0.71 f/EPD 1.75 T23/T56 1.26 BL/EPD 0.42 T23/T78 1.22 TL/EPD 2.20

4th Embodiment

(151) FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment. In FIG. 7, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor **497**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **400**, a first lens element **410**, a second lens element **420**, a third lens element **430**, a fourth lens element **440**, a fifth lens element **450**, a sixth lens element **460**, a seventh lens element **470**, an eighth lens element **480**, an IR-cut filter **490** and an image surface **495**. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element **410** and the second lens element **420**. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element **430**, fourth lens element **440** and the fifth lens element **450**. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element **460**, the seventh lens element **470** and the eighth lens element **480**. The image sensor **497** is disposed on or near the image surface **495** of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (**410-480**). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has negative refractive power, and the third lens group has negative refractive power.

(152) The first lens element **410** with positive refractive power has an object-side surface **411** being convex in a paraxial region thereof and an image-side surface **412** being concave in a paraxial region thereof. The first lens element **410** is made of plastic material and has the object-side surface **411** and the image-side surface **412** being both aspheric.

(153) The second lens element **420** with positive refractive power has an object-side surface **421** being convex in a paraxial region thereof and an image-side surface **422** being convex in a paraxial region thereof. The second lens element **420** is made of plastic material and has the object-side surface **421** and the image-side surface **422** being both aspheric.

(154) The third lens element **430** with negative refractive power has an object-side surface **431** being convex in a paraxial region thereof and an image-side surface **432** being concave in a paraxial region thereof. The third lens element **430** is made of plastic material and has the object-side surface **431** and the image-side surface **432** being both aspheric.

(155) The fourth lens element **440** with positive refractive power has an object-side surface **441** being convex in a paraxial region thereof and an image-side surface **442** being concave in a paraxial region thereof. The fourth lens element **440** is made of plastic material and has the object-side surface **441** and the image-side surface **442** being both aspheric.

(156) The fifth lens element **450** with positive refractive power has an object-side surface **451** being convex in a paraxial region thereof and an image-side surface **452** being convex in a paraxial region thereof. The fifth lens element **450** is made of plastic material and has the object-side surface **451** and the image-side surface **452** being both aspheric.

(157) The sixth lens element **460** with negative refractive power has an object-side surface **461** being concave in a paraxial region thereof and an image-side surface **462** being convex in a paraxial region thereof. The sixth lens element **460** is made of plastic material and has the object-side surface **461** and the image-side surface **462** being both aspheric.

(158) The seventh lens element **470** with positive refractive power has an object-side surface **471**

being convex in a paraxial region thereof and an image-side surface **472** being concave in a paraxial region thereof. The seventh lens element **470** is made of plastic material and has the object-side surface **471** and the image-side surface **472** being both aspheric.

(159) The eighth lens element **480** with negative refractive power has an object-side surface **481** being convex in a paraxial region thereof and an image-side surface **482** being concave in a paraxial region thereof. The eighth lens element **480** is made of plastic material and has the object-side surface **481** and the image-side surface **482** being both aspheric. The image-side surface **482** of the eighth lens element **480** has at least one inflection point.

(160) The IR-cut filter **490** is made of glass and located between the eighth lens element **480** and the image surface **495**, and will not affect the focal length of the photographing optical lens assembly.

(161) The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

(162) TABLE-US-00009 TABLE 7 4th Embodiment $f = 5.98$ mm, $Fno = 1.75$, $HFOV = 33.2$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.631 2 Lens 1 2.663 (ASP) 0.847 Plastic 1.544 55.9 7.05 3 7.722 (ASP)
0.070 4 Lens 2 8.190 (ASP) 0.320 Plastic 1.544 55.9 14.48 5 -203.717 (ASP) 0.040 6 Lens 3 3.272
(ASP) 0.305 Plastic 1.639 23.5 -7.19 7 1.841 (ASP) 0.386 8 Lens 4 6.833 (ASP) 0.856 Plastic
1.544 55.9 22.31 9 14.941 (ASP) 0.071 10 Lens 5 18.422 (ASP) 0.557 Plastic 1.544 55.9 16.48 11
-17.286 (ASP) 0.347 12 Lens 6 -2.939 (ASP) 0.408 Plastic 1.639 23.5 -20.77 13 -3.980 (ASP)
0.038 14 Lens 7 3.405 (ASP) 0.666 Plastic 1.544 55.9 23.22 15 4.339 (ASP) 0.441 16 Lens 8 3.413
(ASP) 0.820 Plastic 1.544 55.9 -13.61 17 2.139 (ASP) 0.550 18 IR-cut filter Plano 0.444 Glass
1.517 64.2 — 19 Plano 0.328 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).

(163) TABLE-US-00010 TABLE 8 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k = 1.6741E-01$
 $-4.8781E+01$ $-8.5057E+01$ $9.0000E+01$ $-3.2961E+01$ $-1.2325E+01$ $A4 = -1.9329E-03$
 $-4.0980E-02$ $-1.0379E-03$ $3.4487E-03$ $-5.3092E-02$ $3.6654E-02$ $A6 = -1.3476E-03$
 $5.0523E-02$ $1.4686E-04$ $4.8289E-05$ $5.8196E-02$ $-3.2300E-02$ $A8 = 3.2383E-03$
 $-3.1314E-02$ — — $-3.3237E-02$ $3.1945E-02$ $A10 = -2.0079E-03$ $1.1208E-02$ — —
 $1.1053E-02$ $-1.6716E-02$ $A12 = 6.0645E-04$ $-2.1005E-03$ — — $-2.1180E-03$ $4.3287E-03$
 $A14 = -7.1566E-05$ $1.4723E-04$ — — $1.5942E-04$ $-4.2417E-04$ Surface # 8 9 10 11 12 13 $k =$
 $-3.4895E+01$ $5.7255E+01$ $9.0000E+01$ $-2.0818E+01$ $6.8347E-01$ $-4.7778E+00$ $A4 =$
 $-1.3773E-03$ $-2.0577E-04$ $-4.7486E-03$ $-3.5896E-02$ $-1.2917E-02$ $-5.1986E-02$ $A6 =$
 $-9.3182E-05$ $-2.6911E-03$ $-3.0559E-03$ $1.7370E-02$ $3.5930E-02$ $3.7822E-02$ $A8 =$
 $1.9786E-03$ $-6.9743E-04$ $-7.5567E-04$ $-6.8386E-03$ $-2.7037E-02$ $-1.9337E-02$ $A10 =$
 $-1.4877E-03$ $-2.1388E-04$ $3.8795E-05$ $-3.3883E-03$ $1.1765E-02$ $6.3981E-03$ $A12 =$
 $3.8512E-04$ $5.0674E-06$ $1.3071E-05$ $4.1256E-03$ $-2.3058E-03$ $-1.0824E-03$ $A14 =$
 $-1.4021E-05$ — — $-1.2572E-03$ $1.4753E-04$ $7.0003E-05$ $A16 =$ — — — $1.2611E-04$ — —
Surface # 14 15 16 17 $k = -1.3541E+01$ $-2.0305E+01$ $-1.0756E+00$ $-1.4857E+00$ $A4 =$
 $-1.2789E-02$ $-3.2998E-03$ $-1.1664E-01$ $-8.6893E-02$ $A6 = 5.0232E-03$ $1.4032E-03$
 $2.6825E-02$ $2.3481E-02$ $A8 = -5.3472E-03$ $-1.6951E-03$ $-3.7535E-03$ $-4.8359E-03$ $A10 =$
 $1.4749E-03$ $3.3299E-04$ $3.9729E-04$ $6.6760E-04$ $A12 = -2.4552E-04$ $-2.3050E-05$
 $-3.0447E-05$ $-5.5999E-05$ $A14 = 1.8979E-05$ $4.7206E-07$ $1.4118E-06$ $2.5393E-06$ $A16 =$
— — $-2.9095E-08$ $-4.7365E-08$

(164) In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

(165) Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

(166) TABLE-US-0001 4th Embodiment f [mm] 5.98 ($\Sigma AT + BL$)/ ΣCT 0.57 Fno 1.75 $TL/ImgH$ 1.88 $HFOV$ [deg.] 33.2 TL [mm] 7.49 $Nmax$ 1.639 $Yc7/f$ (object-side surface 0.24 471) ($V2 + V6$)/ $V1$ 1.42 $Yc7/f$ (image-side surface 0.28 472) ($R11 - R12$)/($R11 + R12$) -0.15 $Yc82/f$ 0.28 ($R15 - R16$)/($R15 + R16$) 0.23 $f/fG1$ 1.21 $f/R16$ 2.79 $f/fG2$ -0.19 $f/f1$ 0.85 $f/fG3$ -0.48 $f1/f2$ 0.49 $f/f7$ 0.26 $Y11/Y82$ 0.51 $f/f8$ -0.44 SD/TD 0.90 $CT3/CT4$ 0.36 $ATmax/ImgH$ 0.11 $CT7/CT8$ 0.81 f/EPD 1.75 $T23/T56$ 0.12 BL/EPD 0.39 $T23/T78$ 0.09 TL/EPD 2.19

5th Embodiment

(167) FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment. In FIG. 9, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 597. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, a seventh lens element 570, an eighth lens element 580, an IR-cut filter 590 and an image surface 595. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 510 and the second lens element 520. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 530, fourth lens element 540 and the fifth lens element 550. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 560, the seventh lens element 570 and the eighth lens element 580. The image sensor 597 is disposed on or near the image surface 595 of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (510-580). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(168) The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being convex in a paraxial region thereof. The first lens element 510 is made of plastic material and has the object-side surface 511 and the image-side surface 512 being both aspheric.

(169) The second lens element 520 with negative refractive power has an object-side surface 521 being concave in a paraxial region thereof and an image-side surface 522 being concave in a paraxial region thereof. The second lens element 520 is made of plastic material and has the object-side surface 521 and the image-side surface 522 being both aspheric.

(170) The third lens element 530 with negative refractive power has an object-side surface 531 being convex in a paraxial region thereof and an image-side surface 532 being concave in a paraxial region thereof. The third lens element 530 is made of plastic material and has the object-side surface 531 and the image-side surface 532 being both aspheric.

(171) The fourth lens element 540 with positive refractive power has an object-side surface 541 being convex in a paraxial region thereof and an image-side surface 542 being concave in a paraxial region thereof. The fourth lens element 540 is made of plastic material and has the object-side surface 541 and the image-side surface 542 being both aspheric.

(172) The fifth lens element 550 with positive refractive power has an object-side surface 551 being convex in a paraxial region thereof and an image-side surface 552 being convex in a paraxial region thereof. The fifth lens element 550 is made of plastic material and has the object-side surface 551 and the image-side surface 552 being both aspheric.

(173) The sixth lens element 560 with negative refractive power has an object-side surface 561 being concave in a paraxial region thereof and an image-side surface 562 being convex in a paraxial region thereof. The sixth lens element 560 is made of plastic material and has the object-

side surface **561** and the image-side surface **562** being both aspheric.

(174) The seventh lens element **570** with positive refractive power has an object-side surface **571** being convex in a paraxial region thereof and an image-side surface **572** being concave in a paraxial region thereof. The seventh lens element **570** is made of plastic material and has the object-side surface **571** and the image-side surface **572** being both aspheric.

(175) The eighth lens element **580** with negative refractive power has an object-side surface **581** being convex in a paraxial region thereof and an image-side surface **582** being concave in a paraxial region thereof. The eighth lens element **580** is made of plastic material and has the object-side surface **581** and the image-side surface **582** being both aspheric. The image-side surface **582** of the eighth lens element **580** has at least one inflection point.

(176) The IR-cut filter **590** is made of glass and located between the eighth lens element **580** and the image surface **595**, and will not affect the focal length of the photographing optical lens assembly.

(177) The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

(178) TABLE-US-00012 TABLE 9 5th Embodiment $f = 5.99$ mm, $Fno = 1.90$, $HFOV = 32.9$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.555 2 Lens 1 2.545 (ASP) 0.842 Plastic 1.544 56.0 4.49 3 -54.860 (ASP)
 0.070 4 Lens 2 -18.590 (ASP) 0.320 Plastic 1.660 20.4 -15.55 5 23.058 (ASP) 0.040 6 Lens 3
 3.822 (ASP) 0.440 Plastic 1.544 56.0 -10.89 7 2.229 (ASP) 0.337 8 Lens 4 6.590 (ASP) 0.327
Plastic 1.544 56.0 21.36 9 14.959 (ASP) 0.238 10 Lens 5 16.790 (ASP) 0.921 Plastic 1.544 56.0
 15.72 11 -17.105 (ASP) 0.209 12 Lens 6 -3.066 (ASP) 0.382 Plastic 1.639 23.5 -10.42 13 -5.964
(ASP) 0.060 14 Lens 7 3.078 (ASP) 0.777 Plastic 1.544 56.0 12.28 15 5.198 (ASP) 0.508 16 Lend
8 3.638 (ASP) 0.816 Plastic 1.544 56.0 -12.14 17 2.160 (ASP) 0.550 18 IR-cut filter Plano 0.444
Glass 1.517 64.2 — 19 Plano 0.227 20 Image Plano — Note: Reference wavelength is 587.6 nm
(d-line).

(179) TABLE-US-00013 TABLE 10 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k =$
 $1.3417E-01$ $-5.0000E+01$ $6.4988E+01$ $9.0000E+01$ $-2.8049E+01$ $-1.4840E+01$ $A4 =$
 $-2.3913E-03$ $-4.2464E-02$ $1.0247E-03$ $1.9540E-03$ $-5.2969E-02$ $3.5248E-02$ $A6 =$
 $-5.9949E-04$ $4.9219E-02$ $1.4687E-03$ $-5.1517E-04$ $5.7058E-02$ $-3.0358E-02$ $A8 =$
 $2.9770E-03$ $-3.0580E-02$ — — $-3.3520E-02$ $3.2003E-02$ $A10 = -1.9813E-03$ $1.1256E-02$
— — $1.0955E-02$ $-1.6646E-02$ $A12 = 6.3177E-04$ $-2.1419E-03$ — — $-2.1251E-03$
 $4.3259E-03$ $A14 = -7.6558E-05$ $1.4780E-04$ — — $1.5469E-04$ $-4.2416E-04$ Surface # 8 9 10
11 12 13 $k = -5.5050E+01$ $5.4815E+01$ $8.7639E+01$ $7.2017E+01$ $8.6604E-01$ $-7.4985E+00$
 $A4 = -2.0686E-03$ $2.2466E-03$ $2.7031E-06$ $-4.0840E-02$ $-1.3755E-02$ $-4.8404E-02$ $A6 =$
 $-4.1153E-04$ $-2.4465E-03$ $-2.4518E-03$ $1.7784E-02$ $3.4786E-02$ $3.8267E-02$ $A8 =$
 $2.1133E-03$ $-7.1953E-04$ $-8.3588E-04$ $-7.0686E-03$ $-2.7068E-02$ $-1.9361E-02$ $A10 =$
 $-1.3173E-03$ $-2.0128E-04$ $-1.7997E-05$ $-3.5305E-03$ $1.1810E-02$ $6.3873E-03$ $A12 =$
 $3.9302E-04$ $5.8034E-05$ $-4.1862E-05$ $4.0983E-03$ $-2.3031E-03$ $-1.0822E-03$ $A14 =$
 $-1.7043E-05$ — — $-1.2569E-03$ $1.4272E-04$ $7.0849E-05$ $A16 =$ — — — $1.2661E-04$ — —
Surface # 14 15 16 17 $k = -1.4423E+01$ $-4.1659E+01$ $-9.6976E-01$ $-1.1592E+00$ $A4 =$
 $-1.3553E-02$ $-3.4495E-03$ $-1.1615E-01$ $-8.8977E-02$ $A6 = 4.9109E-03$ $1.3387E-03$
 $2.6785E-02$ $2.3550E-02$ $A8 = -5.3404E-03$ $-1.7104E-03$ $-3.7550E-03$ $-4.8319E-03$ $A10 =$
 $1.4639E-03$ $3.3214E-04$ $3.9737E-04$ $6.6770E-04$ $A12 = -2.3980E-04$ $-2.3059E-05$
 $-3.0443E-05$ $-5.5974E-05$ $A14 = 1.9746E-05$ $4.7762E-07$ $1.4139E-06$ $2.5390E-06$ $A16 =$
— — $-2.8805E-08$ $-4.7469E-08$

(180) In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided

again.

(181) Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

(182) TABLE-US-00014 5th Embodiment f [mm] 5.99 $(\Sigma AT + BL)/\Sigma CT$ 0.56 Fno 1.90 $TL/ImgH$ 1.88 $HFOV$ [deg.] 32.9 TL [mm] 7.51 $Nmax$ 1.660 $Yc7/f$ (object-side surface 0.23 571) $(V2 + V6)/V1$ 0.78 $Yc7/f$ (image-side surface 0.26 572) $(R11 - R12)/(R11 + R12)$ -0.32 $Yc82/f$ 0.26 $(R15 - R16)/(R15 + R16)$ 0.25 $f/fG1$ 1.01 $f/R16$ 2.77 $f/fG2$ 0.11 $f/f1$ 1.33 $f/fG3$ -0.61 $f1/f2$ -0.29 $f/f7$ 0.49 $Y11/Y82$ 0.46 $f/f8$ -0.49 SD/TD 0.91 $CT3/CT4$ 1.35 $ATmax/ImgH$ 0.13 $CT7/CT8$ 0.95 f/EPD 1.90 $T23/T56$ 0.19 BL/EPD 0.39 $T23/T78$ 0.08 TL/EPD 2.38

6th Embodiment

(183) FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment. In FIG. 11, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 690. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, a seventh lens element 670, an eighth lens element 680, an IR-cut filter 690 and an image surface 695. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 610 and the second lens element 620. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 630, fourth lens element 640 and the fifth lens element 650. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 660, the seventh lens element 670 and the eighth lens element 680. The image sensor 697 is disposed on or near the image surface 695 of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (610-680). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(184) The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof. The sixth lens element 610 is made of plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric.

(185) The second lens element 620 with negative refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being concave in a paraxial region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric.

(186) The third lens element 630 with positive refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being convex in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

(187) The fourth lens element 640 with negative refractive power has an object-side surface 641 being concave in a paraxial region thereof and an image-side surface 642 being concave in a paraxial region thereof. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric.

(188) The fifth lens element 650 with positive refractive power has an object-side surface 651 being convex in a paraxial region thereof and an image-side surface 652 being convex in a paraxial region thereof. The fifth lens element 650 is made of plastic material and has the object-side surface 651 and the image-side surface 652 being both aspheric.

(189) The sixth lens element **660** with positive refractive power has an object-side surface **661** being concave in a paraxial region thereof and an image-side surface **662** being convex in a paraxial region thereof. The sixth lens element **660** is made of plastic material and has the object-side surface **661** and the image-side surface **662** being both aspheric.

(190) The seventh lens element **670** with negative refractive power has an object-side surface **671** being concave in a paraxial region thereof and an image-side surface **672** being concave in a paraxial region thereof. The seventh lens element **670** is made of plastic material and has the object-side surface **671** and the image-side surface **672** being both aspheric.

(191) The eighth lens element **680** with negative refractive power has an object-side surface **681** being convex in a paraxial region thereof and an image-side surface **682** being concave in a paraxial region thereof. The eighth lens element **680** is made of plastic material and has the object-side surface **681** and the image-side surface **682** being both aspheric. The image-side surface **682** of the eighth lens element **680** has at least one inflection point.

(192) The IR-cut filter **690** is made of glass and located between the eighth lens element **680** and the image surface **695**, and will not affect the focal length of the photographing optical lens assembly.

(193) The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

(194) TABLE-US-00015 TABLE 11 6th Embodiment $f = 4.37$ mm, $Fno = 1.95$, $HFOV = 32.7$ deg.

Surface #	Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length	0	Object	Plano	Infinity		
1	Ape. Stop	Plano	-0.401	2	Lens 1	1.705 (ASP)	0.692	Plastic	1.544	55.9	3.30	3	28.524 (ASP)
0.070	4	Lens 2	2.874 (ASP)	0.230	Plastic	1.639	23.5	-5.68	5	1.554 (ASP)	0.250	6	Lens 3
(ASP)	0.459	Plastic	1.544	55.9	3.30	7	-11.214 (ASP)	0.053	8	Lens 4	-9.262 (ASP)	0.151	Plastic
1.544	55.9	-13.95	9	42.304 (ASP)	0.205	10	Lens 5	62.900 (ASP)	0.443	Plastic	1.544	55.9	17.07
11	-10.872 (ASP)	0.134	12	Lens 6	-2.665 (ASP)	0.314	Plastic	1.639	23.5	142.31	13	-2.708 (ASP)	
0.121	14	Lens 7	-30.102 (ASP)	0.454	Plastic	1.544	55.9	-17.47	15	13.962 (ASP)	0.126	16	Lend 8
2.066 (ASP)	0.514	Plastic	1.544	55.9	-9.88	17	1.361 (ASP)	0.450	18	IR-cut filter	Plano	0.210	
Glass	1.517	64.2	—	19	Plano	0.301	20	Image	Plano	—	Note: Reference wavelength is 587.6 nm (d-line).		

(195) TABLE-US-00016 TABLE 12 Aspheric Coefficients

Surface #	2	3	4	5	6	7	k =				
2.1674E-01	-4.9516E+01	-3.7522E+01	-1.3303E+01	-1.9999E+01	8.5093E+01	A4 =					
1.3946E-03	-1.3323E-01	-1.7545E-01	1.2855E-01	-6.3965E-03	-9.5283E-03	A6 =					
-2.0598E-02	3.6306E-01	4.0584E-01	-2.3141E-01	-1.9316E-03	-6.5486E-03	A8 =					
5.0644E-02	-4.9238E-01	-5.1185E-01	4.8795E-01	3.4843E-02	3.3721E-03	A10 =					
-6.7715E-02	3.8057E-01	3.7933E-01	-5.4822E-01	-4.2147E-02	—	A12 =	4.5003E-02				
-1.5437E-01	-1.5563E-01	3.2179E-01	2.6549E-02	—	A14 =	-1.1705E-02	2.4066E-02				
2.6062E-02	-6.9334E-02	-2.2918E-03	—	Surface #	8	9	10	11	12	13	k =
8.9677E+01	-8.7044E+01	6.5604E+01	1.6291E+00	-1.2743E+01	A4 =	-3.1962E-03					
-1.8814E-02	-7.0542E-02	-9.3010E-02	-8.9983E-02	-1.6441E-01	A6 =	-3.3006E-03					
-2.4056E-02	-3.4329E-02	9.2570E-02	2.6421E-01	2.5430E-01	A8 =	-7.9705E-03					
-1.9487E-03	-2.8803E-02	-1.0356E-01	-4.2611E-01	-3.0319E-01	A10 =	—	-8.1491E-03				
4.8180E-03	-1.1588E-01	3.9726E-01	2.1822E-01	A12 =	—	1.4562E-04	8.8153E-07				
3.0818E-01	-1.7202E-01	-8.0505E-02	A14 =	—	—	-2.0573E-01	2.4455E-02				
1.1567E-02	A16 =	—	—	—	4.5154E-02	—	—				
Surface #	14	15	16	17	k =	-9.0000E+01					
-8.9993E+01	-9.0284E-01	-2.0562E+00	A4 =	2.7669E-02	-1.4887E-02	-3.7958E-01					
-2.7011E-01	A6 =	-2.0492E-02	1.4428E-02	1.8820E-01	1.6545E-01	A8 =	-7.0510E-02				
-2.6799E-02	-5.8286E-02	-7.5297E-02	A10 =	5.2817E-02	1.1156E-02	1.3553E-02					
2.2762E-02	A12 =	-1.8630E-02	-1.7202E-03	-2.2705E-03	-4.1759E-03	A14 =	2.8558E-03				
8.3097E-05	2.3095E-04	4.1513E-04	A16 =	—	—	-1.0359E-05	-1.7022E-05				

(196) In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned

lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

(197) Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

(198) TABLE-US-00017 6th Embodiment f [mm] 4.37 ($\Sigma AT + BL$)/ ΣCT 0.59 Fno 1.95 $TL/ImgH$ 1.79 $HFOV$ [deg.] 32.7 TL [mm] 5.18 $Nmax$ 1.639 $Yc7/f$ (object-side surface — 671) ($V2 + V6$)/ $V1$ 0.84 $Yc7/f$ (image-side surface 0.20 672) ($R11 - R12$)/($R11 + R12$) -0.01 $Yc82/f$ 0.26 ($R15 - R16$)/($R15 + R16$) 0.21 $f/fG1$ 0.76 $f/R16$ 3.21 $f/fG2$ 0.56 $f/f1$ 1.32 $f/fG3$ -0.73 $f1/f2$ -0.58 $f/f7$ -0.25 $Y11/Y82$ 0.48 $f/f8$ -0.44 SD/TD 0.90 $CT3/CT4$ 3.04 $ATmax/ImgH$ 0.09 $CT7/CT8$ 0.88 f/EPD 1.95 $T23/T56$ 1.87 BL/EPD 0.43 $T23/T78$ 1.98 TL/EPD 2.31

7th Embodiment

(199) FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure. FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 7th embodiment. In FIG. 13, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 797. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, a seventh lens element 770, an eighth lens element 780, an IR-cut filter 790 and an image surface 795. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 710 and the second lens element 720. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 730, fourth lens element 740 and the fifth lens element 750. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 760, the seventh lens element 770 and the eighth lens element 780. The image sensor 797 is disposed on or near the image surface 795 of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (710-780). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has positive refractive power.

(200) The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof. The first lens element 710 is made of plastic material and has the object-side surface 711 and the image-side surface 712 being both aspheric.

(201) The second lens element 720 with negative refractive power has an object-side surface 721 being convex in a paraxial region thereof and an image-side surface 722 being concave in a paraxial region thereof. The second lens element 720 is made of plastic material and has the object-side surface 721 and the image-side surface 722 being both aspheric.

(202) The third lens element 730 with positive refractive power has an object-side surface 731 being convex in a paraxial region thereof and an image-side surface 732 being concave in a paraxial region thereof. The third lens element 730 is made of plastic material and has the object-side surface 731 and the image-side surface 732 being both aspheric.

(203) The fourth lens element 740 with positive refractive power has an object-side surface 741 being convex in a paraxial region thereof and an image-side surface 742 being convex in a paraxial region thereof. The fourth lens element 740 is made of plastic material and has the object-side surface 741 and the image-side surface 742 being both aspheric.

(204) The fifth lens element 750 with negative refractive power has an object-side surface 751

being concave in a paraxial region thereof and an image-side surface 752 being convex in a paraxial region thereof. The fifth lens element 750 is made of plastic material and has the object-side surface 751 and the image-side surface 752 being both aspheric.

(205) The sixth lens element 760 with positive refractive power has an object-side surface 761 being convex in a paraxial region thereof and an image-side surface 762 being convex in a paraxial region thereof. The sixth lens element 760 is made of plastic material and has the object-side surface 761 and the image-side surface 762 being both aspheric.

(206) The seventh lens element 770 with positive refractive power has an object-side surface 771 being concave in a paraxial region thereof and an image-side surface 772 being convex in a paraxial region thereof. The seventh lens element 770 is made of plastic material and has the object-side surface 771 and the image-side surface 772 being both aspheric.

(207) The eighth lens element 780 with negative refractive power has an object-side surface 781 being concave in a paraxial region thereof and an image-side surface 782 being concave in a paraxial region thereof. The eighth lens element 780 is made of plastic material and has the object-side surface 781 and the image-side surface 782 being both aspheric. The image-side surface 782 of the eighth lens element 780 has at least one inflection point.

(208) The IR-cut filter 790 is made of glass and located between the eighth lens element 780 and the image surface 795, and will not affect the focal length of the photographing optical lens assembly.

(209) The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

(210) TABLE-US-00018 TABLE 13 7th Embodiment $f = 6.35$ mm, $Fno = 2.20$, $HFOV = 42.0$ deg.

Surface #	Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length	0	Object	Plano	Infinity
1	Ape. Stop	Plano	-0.331	2 Lens 1	3.431 (ASP)	0.718	Plastic	1.544	55.9	10.41	3 8.057 (ASP)
0.188	4 Lens 2	4.716 (ASP)	0.396	Plastic	1.640	23.3	-21.74	5 3.406 (ASP)	0.355	6 Lens 3	8.194 (ASP)
0.588	Plastic	1.544	55.9	20.96	7 28.361 (ASP)	0.263	8 Lens 4	12.385 (ASP)	0.699	Plastic	1.544
55.9	14.42	9 -21.007 (ASP)	0.568	10 Lens 5	-1.734 (ASP)	0.411	Plastic	1.640	23.3	-9.30	11 -2.674 (ASP)
0.072	12 Lens 6	5.765 (ASP)	0.887	Plastic	1.544	55.9	8.22	13 -18.931 (ASP)	0.336	14 Lens 7	-10.088 (ASP)
0.690	Plastic	1.544	55.9	4.56	15 -2.039 (ASP)	0.362	16 Lend 8	-38.051 (ASP)	0.745	Plastic	1.530
55.8	-2.88	17 1.601 (ASP)	1.050	18 IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	19 Plano
0.320	20 Image	Plano	—	Note: Reference wavelength is 587.6 nm (d-line).							

(211) TABLE-US-00019 TABLE 14 Aspheric Coefficients

Surface #	2	3	4	5	6	7	k =
8.9915E-01	1.0446E+01	-9.7945E+00	-5.7630E+00	-3.4397E+00	-7.5106E+00	A4 =	
2.1862E-04	-1.3002E-02	-2.8750E-02	-1.4489E-02	-1.5964E-03	-8.1781E-03	A6 =	
9.3154E-04	8.4048E-03	1.2076E-02	6.3792E-03	-1.6598E-04	6.2768E-04	A8 =	
3.7742E-04	-1.7322E-03	-2.9274E-03	-1.2331E-03	-1.4227E-06	-6.3861E-05	A10 =	
-4.6373E-04	3.2602E-06	7.4904E-05	1.3194E-04	-6.5983E-05	-5.5516E-05	A12 =	
2.4135E-04	8.1796E-05	1.0317E-04	-6.1925E-05	—	—	A14 =	-3.4857E-05
-4.1887E-06	-2.7815E-05	1.1831E-05	—	—	—	Surface #	8 9 10 11 12 13
k =	-4.9161E+01	3.0000E+00	-2.1379E+00	-3.9420E+00	1.0633E+00	2.9486E+00	A4 = -1.4353E-02
-9.4664E-03	3.4117E-02	1.1014E-02	-1.5491E-02	1.3464E-03	A6 = -1.1328E-03	-2.5409E-03	-1.5949E-02
-5.0593E-03	2.7379E-03	-2.5908E-04	A8 = 3.5833E-04	1.9952E-04	4.0038E-03	1.1742E-03	-3.7208E-04
-2.5237E-06	A10 = -5.6257E-05	-2.1845E-06	-6.7402E-04	-1.7557E-04	1.5345E-05	-6.8350E-08	A12 = -1.5438E-05
-9.3498E-07	8.7207E-05	1.9021E-05	5.3033E-07	—	A14 = 3.0947E-06	4.4233E-07	-5.6319E-06
-9.1764E-07	-8.4643E-08	—	Surface #	14 15 16 17	k = 2.6774E+00	-6.7965E+00	8.5000E+01
-6.5954E+00	A4 = -1.9919E-03	-4.8403E-03	-1.5278E-02	-6.6138E-03	A6 = 1.0979E-04	-1.1705E-04	7.1418E-04
4.5320E-04	A8 = 1.8379E-06	3.8093E-04	-2.7403E-06	-2.5975E-05	A10 = -1.6074E-07	-6.1316E-05	3.3653E-07
8.4289E-07	A12 =						

— 3.7673E-06 8.0297E-09 -1.7531E-08 A14 = — -8.2877E-08 -1.6049E-09 1.8760E-10
(212) In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

(213) Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

(214) TABLE-US-00020 7th Embodiment f [mm] 6.35 ($\Sigma AT + BL$)/ ΣCT 0.74 Fno 2.20 TL/ImgH 1.49 HFOV [deg.] 42.0 TL [mm] 8.95 Nmax 1.640 Yc7/f(object-side surface — 771) (V2 + V6)/V1 1.42 Yc7/f(image-side surface — 772) (R11 - R12)/(R11 + R12) -1.88 Yc82/f 0.49 (R15 - R16)/(R15 + R16) 1.09 f/fG1 0.37 f/R16 3.97 f/fG2 0.09 f/f1 0.61 f/fG3 0.57 f1/f2 -0.48 f/f7 1.39 Y11/Y82 0.27 f/f8 -2.20 SD/TD 0.95 CT3/CT4 0.84 ATmax/ImgH 0.09 CT7/CT8 0.93 f/EPD 2.20 T23/T56 4.93 BL/EPD 0.58 T23/T78 0.98 TL/EPD 3.10

8th Embodiment

(215) FIG. 15 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure. FIG. 16 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 8th embodiment. In FIG. 15, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 897. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 800, a first lens element 810, a second lens element 820, a third lens element 830, a fourth lens element 840, a fifth lens element 850, a sixth lens element 860, a seventh lens element 870, an eighth lens element 880, an IR-cut filter 890 and an image surface 895. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 810 and the second lens element 820. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 830, fourth lens element 840 and the fifth lens element 850. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 860, the seventh lens element 870 and the eighth lens element 880. The image sensor 897 is disposed on or near the image surface 895 of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (810-880). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has positive refractive power.

(216) The first lens element 810 with positive refractive power has an object-side surface 811 being convex in a paraxial region thereof and an image-side surface 812 being concave in a paraxial region thereof. The first lens element 810 is made of plastic material and has the object-side surface 811 and the image-side surface 812 being both aspheric.

(217) The second lens element 320 with negative refractive power has an object-side surface 821 being convex in a paraxial region thereof and an image-side surface 822 being concave in a paraxial region thereof. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being both aspheric.

(218) The third lens element 830 with positive refractive power has an object-side surface 831 being convex in a paraxial region thereof and an image-side surface 832 being concave in a paraxial region thereof. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being both aspheric.

(219) The fourth lens element 840 with positive refractive power has an object-side surface 841 being convex in a paraxial region thereof and an image-side surface 842 being convex in a paraxial region thereof. The fourth lens element 840 is made of plastic material and has the object-side

surface **841** and the image-side surface **842** being both aspheric.

(220) The fifth lens element **850** with negative refractive power has an object-side surface **851** being concave in a paraxial region thereof and an image-side surface **852** being convex in a paraxial region thereof. The fifth lens element **850** is made of plastic material and has the object-side surface **851** and the image-side surface **852** being both aspheric.

(221) The sixth lens element **860** with positive refractive power has an object-side surface **861** being convex in a paraxial region thereof and an image-side surface **862** being concave in a paraxial region thereof. The sixth lens element **860** is made of plastic material and has the object-side surface **861** and the image-side surface **862** being both aspheric.

(222) The seventh lens element **870** with positive refractive power has an object-side surface **871** being convex in a paraxial region thereof and an image-side surface **872** being convex in a paraxial region thereof. The seventh lens element **870** is made of plastic material and has the object-side surface **871** and the image-side surface **872** being both aspheric.

(223) The eighth lens element **880** with negative refractive power has an object-side surface **881** being concave in a paraxial region thereof and an image-side surface **882** being concave in a paraxial region thereof. The eighth lens element **880** is made of plastic material and has the object-side surface **881** and the image-side surface **882** being both aspheric. The image-side surface **882** of the eighth lens element **880** has at least one inflection point.

(224) The IR-cut filter **890** is made of glass and located between the eighth lens element **880** and the image surface **895**, and will not affect the focal length of the photographing optical lens assembly.

(225) The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

(226) TABLE-US-00021 TABLE 15 8th Embodiment $f = 6.05$ mm, $Fno = 2.50$, $HFOV = 38.8$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.230 2 Lens 1 3.225 (ASP) 0.789 Plastic 1.544 55.9 9.63 3 7.663 (ASP)
 0.168 4 Lens 2 4.692 (ASP) 0.346 Plastic 1.640 23.3 -18.10 5 3.243 (ASP) 0.314 6 Lens 3 5.692
(ASP) 0.476 Plastic 1.544 55.9 19.25 7 12.098 (ASP) 0.460 8 Lens 4 14.559 (ASP) 0.724 Plastic
 1.544 55.9 9.46 9 -7.823 (ASP) 0.353 10 Lens 5 -1.743 (ASP) 0.289 Plastic 1.640 23.3 -6.66 11
 -3.140 (ASP) 0.226 12 Lens 6 6.722 (ASP) 0.364 Plastic 1.640 23.3 20.62 13 13.416 (ASP) 0.102
14 Lens 7 17.648 (ASP) 0.972 Plastic 1.544 55.9 3.40 15 -2.025 (ASP) 0.432 16 Lens 8 -17.447
(ASP) 0.788 Plastic 1.530 55.8 -2.80 17 1.646 (ASP) 1.050 18 IR-cut filter Plano 0.316 Glass
 1.517 64.2 — 19 Plano 0.251 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).

(227) TABLE-US-00022 TABLE 16 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k =$
 $9.7327E-01$ $1.3914E+01$ $-1.0179E+01$ $-5.7767E+00$ $-5.2840E+00$ $-1.2252E+01$ $A4 =$
 $2.7811E-04$ $-1.0851E-02$ $-2.8930E-02$ $-1.4065E-02$ $-1.9697E-03$ $-8.8286E-03$ $A6 =$
 $1.0603E-03$ $8.5807E-03$ $1.2086E-02$ $6.4812E-03$ $2.4629E-04$ $6.5023E-04$ $A8 =$
 $3.3103E-04$ $-1.7220E-03$ $-3.0661E-03$ $-1.2735E-03$ $1.5592E-04$ $-1.3966E-04$ $A10 =$
 $-4.2729E-04$ $1.2047E-04$ $3.4757E-05$ $8.4359E-05$ $-6.5196E-05$ $-6.0865E-05$ $A12 =$
 $3.2854E-04$ $1.4342E-04$ $9.7425E-05$ $-8.2394E-05$ — — $A14 = -6.0799E-05$ $2.8783E-05$
 $2.3134E-06$ $1.6483E-05$ — — Surface # 8 9 10 11 12 13 $k = -2.6542E+01$ $-1.6034E+00$
 $-2.1244E+00$ $-4.7602E+00$ $2.1182E+00$ $-2.0000E+01$ $A4 = -1.4866E-02$ $-1.0663E-02$
 $3.4762E-02$ $1.0584E-02$ $-1.4918E-02$ $-3.5137E-04$ $A6 = -1.5743E-03$ $-2.4323E-03$
 $-1.5850E-02$ $-5.1466E-03$ $2.5301E-03$ $-3.6027E-04$ $A8 = 3.4251E-04$ $2.1082E-04$
 $4.0076E-03$ $1.1666E-03$ $-3.8897E-04$ $-1.1243E-05$ $A10 = -5.0557E-05$ $-1.1461E-06$
 $-6.7647E-04$ $-1.7611E-04$ $1.7863E-05$ $-1.5867E-06$ $A12 = -1.5725E-05$ $-2.9170E-08$
 $8.5997E-05$ $1.9016E-05$ $6.7620E-07$ — $A14 = 2.1634E-06$ $8.8760E-07$ $-6.0050E-06$
 $-9.2073E-07$ $-1.2743E-07$ — Surface # 14 15 16 17 $k = -2.0000E+01$ $-6.6776E+00$
 $2.8072E+00$ $-6.8316E+00$ $A4 = -4.3273E-03$ $-2.2862E-03$ $-1.5713E-02$ $-6.8908E-03$ $A6 =$
 $4.4821E-05$ $-1.6718E-04$ $7.4870E-04$ $4.7037E-04$ $A8 = 5.3925E-07$ $3.7849E-04$

-1.3310E-06 -2.6480E-05 A10 = -1.2113E-07 -6.1384E-05 3.5705E-07 8.0096E-07 A12 =
— 3.7687E-06 4.1296E-09 -1.7720E-08 A14 = — -8.2605E-08 -1.9486E-09 2.6140E-10
(228) In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

(229) Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

(230) TABLE-US-00023 8th Embodiment f [mm] 6.05 ($\Sigma AT + BL$)/ ΣCT 0.77 Fno 2.50 TL/ImgH 1.65 HFOV [deg.] 38.8 TL [mm] 8.42 Nmax 1.640 Yc7/f(object-side surface 0.29 871) (V2 + V6)/V1 0.83 Yc7/f(image-side surface — 872) (R11 - R12)/(R11 + R12) -0.33 Yc82/f 0.50 (R15 - R16)/(R15 + R16) 1.21 f/fG1 0.35 f/R16 3.67 f/fG2 0.10 f/f1 0.63 f/fG3 0.61 f1/f2 -0.53 f/f7 1.78 Y11/Y82 0.28 f/f8 -2.16 SD/TD 0.97 CT3/CT4 0.66 ATmax/ImgH 0.09 CT7/CT8 1.23 f/EPD 2.50 T23/T56 1.39 BL/EPD 0.67 T23/T78 0.73 TL/EPD 3.48

9th Embodiment

(231) FIG. 17 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure. FIG. 18 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 9th embodiment. In FIG. 17, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 997. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 900, a first lens element 910, a second lens element 920, a third lens element 930, a fourth lens element 940, a fifth lens element 950, a sixth lens element 960, a seventh lens element 970, an eighth lens element 980, an IR-cut filter 990 and an image surface 995. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 910 and the second lens element 920. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 930, fourth lens element 940 and the fifth lens element 950. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 960, the seventh lens element 970 and the eighth lens element 980. The image sensor 997 is disposed on or near the image surface 995 of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (910-980). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(232) The first lens element 910 with positive refractive power has an object-side surface 911 being convex in a paraxial region thereof and an image-side surface 912 being concave in a paraxial region thereof. The first lens element 910 is made of plastic material and has the object-side surface 911 and the image-side surface 912 being both aspheric.

(233) The second lens element 920 with negative refractive power has an object-side surface 921 being convex in a paraxial region thereof and an image-side surface 922 being concave in a paraxial region thereof. The second lens element 920 is made of plastic material and has the object-side surface 921 and the image-side surface 922 being both aspheric.

(234) The third lens element 930 with positive refractive power has an object-side surface 931 being convex in a paraxial region thereof and an image-side surface 932 being convex in a paraxial region thereof. The third lens element 930 is made of plastic material and has the object-side surface 931 and the image-side surface 932 being both aspheric.

(235) The fourth lens element 940 with negative refractive power has an object-side surface 941 being concave in a paraxial region thereof and an image-side surface 942 being concave in a

paraxial region thereof. The fourth lens element **940** is made of plastic material and has the object-side surface **941** and the image-side surface **942** being both aspheric.

(236) The fifth lens element **950** with positive refractive power has an object-side surface **951** being concave in a paraxial region thereof and an image-side surface **952** being convex in a paraxial region thereof. The fifth lens element **950** is made of plastic material and has the object-side surface **951** and the image-side surface **952** being both aspheric.

(237) The sixth lens element **960** with positive refractive power has an object-side surface **961** being concave in a paraxial region thereof and an image-side surface **962** being convex in a paraxial region thereof. The sixth lens element **960** is made of plastic material and has the object-side surface **961** and the image-side surface **962** being both aspheric.

(238) The seventh lens element **970** with negative refractive power has an object-side surface **971** being convex in a paraxial region thereof and an image-side surface **972** being concave in a paraxial region thereof. The fifth lens element **970** is made of plastic material and has the object-side surface **971** and the image-side surface **972** being both aspheric.

(239) The eighth lens element **980** with negative refractive power has an object-side surface **981** being convex in a paraxial region thereof and an image-side surface **982** being concave in a paraxial region thereof. The eighth lens element **980** is made of plastic material and has the object-side surface **981** and the image-side surface **982** being both aspheric. The image-side surface **982** of the eighth lens element **980** has at least one inflection point.

(240) The IR-cut filter **990** is made of glass and located between the eighth lens element **980** and the image surface **995**, and will not affect the focal length of the photographing optical lens assembly.

(241) The detailed optical data of the 9th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

(242) TABLE-US-00024 TABLE 17 9th Embodiment $f = 6.35$ mm, $Fno = 1.80$, $HFOV = 34.4$ deg.

Surface #	Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length	0	Object	Plano	Infinity	
1	Ape. Stop	Plano	-0.698	2 Lens 1	2.628 (ASP)	0.912	Plastic	1.544	55.9	5.23	3	30.092 (ASP)
0.110	4 Lens 2	4.868 (ASP)	0.408	Plastic	1.639	23.5	-9.27	5	2.584 (ASP)	0.432	6	Lens 3
(ASP)	0.611	Plastic	1.544	55.9	11.09	7	-17.258 (ASP)	0.080	8	Lens 4	-14.763 (ASP)	0.243
Plastic	1.544	55.9	-24.38	9	131.813 (ASP)	0.237	10	Lens 5	-408.163 (ASP)	0.679	Plastic	1.544
55.9	31.63	11	-16.526 (ASP)	0.336	12	Lens 6	-3.215 (ASP)	0.390	Plastic	1.639	23.5	75.00
13	-3.155 (ASP)	0.035	14	Lens 7	4.465 (ASP)	0.623	Plastic	1.544	55.9	-34.61	15	3.432 (ASP)
0.553	16	Lens 8	4.324 (ASP)	0.838	Plastic	1.544	55.9	-12.29	17	2.446 (ASP)	0.600	18
IR-cut filter	Plano	0.326	Glass	1.517	64.2	—	19	Plano	0.282	20	Image	Plano
—	Note:	Reference wavelength is	587.6 nm (d-line).									

(243) TABLE-US-00025 TABLE 18 Aspheric Coefficients

Surface #	2	3	4	5	6	7	k =			
1	1.8867E-01	-5.0000E+01	-3.7126E+01	-1.4019E+01	-2.2856E+01	8.9705E+01	A4 =			
1	1.0690E-03	-3.7048E-02	-4.5877E-02	3.5651E-02	-3.9792E-03	-4.6341E-03	A6 =			
-2.3224E-03	4.0543E-02	4.4753E-02	-2.5798E-02	-1.3863E-03	-8.5797E-04	A8 =				
2.3501E-03	-2.2626E-02	-2.3596E-02	2.2418E-02	1.5124E-03	—	A10 =	-1.2612E-03			
7.2857E-03	7.3388E-03	-1.0521E-02	-8.6274E-04	—	A12 =	3.4896E-04	-1.2297E-03			
-1.2243E-03	2.6180E-03	1.8949E-04	—	A14 =	-3.7428E-05	8.0213E-05	7.9397E-05			
-2.3333E-04	-4.3104E-06	—	Surface #	8	9	10	11	12	13	k =
3.9077E+01	-9.0000E+01	6.5040E+01	1.3817E+00	-1.6373E+01	A4 =	-2.2212E-03	-5.5460E-03			
-1.7005E-02	-3.3633E-02	-9.2720E-03	-4.8459E-02	A6 =	-3.6569E-04	-4.4737E-03				
-4.7753E-03	1.2586E-02	2.7049E-02	2.8738E-02	A8 =	—	-1.5494E-04	-1.5784E-03			
-4.8008E-03	-1.9676E-02	-1.3990E-02	A10 =	—	-7.4992E-05	1.0910E-04	-2.2590E-03			
7.7222E-03	4.1663E-03	A12 =	—	2.0351E-06	6.8999E-05	2.4441E-03	-1.3588E-03			
-6.3966E-04	A14 =	—	—	-6.7819E-04	7.9979E-05	3.8742E-05	A16 =			
—	—	—	—	—	—	—	—			
6.2555E-05	—	—	Surface #	14	15	16	17	k =		
-1.0191E+01	-2.5236E+01	-7.5301E-01								

-1.1325E+00 A4 = -1.7204E-02 -6.0295E-03 -1.0068E-01 -7.7178E-02 A6 = -4.4444E-03
1.4208E-03 2.1108E-02 1.8417E-02 A8 = -3.9107E-03 -1.2160E-03 -2.6955E-03
-3.4654E-03 A10 = 9.5751E-04 2.1390E-04 2.5898E-04 4.3552E-04 A12 = -1.4308E-04
-1.3750E-05 -1.8057E-05 -3.3201E-05 A14 = 1.0003E-05 2.7122E-07 7.6245E-07
1.3695E-06 A16 = 0.0000E+00 — -1.4060E-08 -2.3245E-08

(244) In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

(245) Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following conditions:

(246) TABLE-US-00026 9th Embodiment f [mm] 6.35 ($\Sigma AT + BL$)/ ΣCT 0.64 Fno 1.80 TL/ImgH 1.71 HFOV [deg.] 34.4 TL [mm] 7.69 Nmax 1.639 Yc7/f(object-side surface 0.22 971) (V2 + V6)/V1 0.84 Yc7/f(image-side surface 0.26 972) (R11 - R12)/(R11 + R12) 0.01 Yc82/f 0.26 (R15 - R16)/(R15 + R16) 0.28 f/fG1 0.70 f/R16 2.59 f/fG2 0.51 f/f1 1.21 f/fG3 -0.63 f1/f2 -0.56 f/f7 -0.18 Y11/Y82 0.47 f/f8 -0.52 SD/TD 0.89 CT3/CT4 2.51 ATmax/ImgH 0.12 CT7/CT8 0.74 f/EPD 1.80 T23/T56 1.29 BL/EPD 0.34 T23/T78 0.78 TL/EPD 2.18

10th Embodiment

(247) FIG. 19 is a schematic view of an image capturing unit according to the 10th embodiment of the present disclosure. FIG. 20 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 10th embodiment. In FIG. 19, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 1090. The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element 1010, an aperture stop 1000, a second lens element 1020, a third lens element 1030, a fourth lens element 1040, a fifth lens element 1050, a sixth lens element 1060, a seventh lens element 1070, an eighth lens element 1080, an IR-cut filter 1090 and an image surface 1095. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 1010 and the second lens element 1020. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 1030, fourth lens element 1040 and the fifth lens element 1050. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 1060, the seventh lens element 1070 and the eighth lens element 1080. The image sensor 1097 is disposed on or near the image surface 1095 of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (1010-1080). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(248) The first lens element 1010 with positive refractive power has an object-side surface 1011 being convex in a paraxial region thereof and an image-side surface 1012 being convex in a paraxial region thereof. The first lens element 1010 is made of plastic material and has the object-side surface 1011 and the image-side surface 1012 being both aspheric.

(249) The second lens element 1020 with negative refractive power has an object-side surface 1021 being convex in a paraxial region thereof and an image-side surface 1022 being concave in a paraxial region thereof. The second lens element 1020 is made of plastic material and has the object-side surface 1021 and the image-side surface 1022 being both aspheric.

(250) The third lens element 1030 with negative refractive power has an object-side surface 1031 being concave in a paraxial region thereof and an image-side surface 1032 being concave in a paraxial region thereof. The third lens element 1030 is made of plastic material and has the object-

side surface **1031** and the image-side surface **1032** being both aspheric.

(251) The fourth lens element **1040** with positive refractive power has an object-side surface **1041** being convex in a paraxial region thereof and an image-side surface **1042** being convex in a paraxial region thereof. The fourth lens element **1040** is made of plastic material and has the object-side surface **1041** and the image-side surface **1042** being both aspheric.

(252) The fifth lens element **1050** with positive refractive power has an object-side surface **1051** being convex in a paraxial region thereof and an image-side surface **1052** being concave in a paraxial region thereof. The fifth lens element **1050** is made of plastic material and has the object-side surface **1051** and the image-side surface **1052** being both aspheric.

(253) The sixth lens element **1060** with positive refractive power has an object-side surface **1061** being concave in a paraxial region thereof and an image-side surface **1062** being convex in a paraxial region thereof. The sixth lens element **1060** is made of plastic material and has the object-side surface **1061** and the image-side surface **1062** being both aspheric.

(254) The seventh lens element **1070** with negative refractive power has an object-side surface **1071** being convex in a paraxial region thereof and an image-side surface **1072** being concave in a paraxial region thereof. The seventh lens element **1070** is made of plastic material and has the object-side surface **1071** and the image-side surface **1072** being both aspheric.

(255) The eighth lens element **1080** with negative refractive power has an object-side surface **1081** being convex in a paraxial region thereof and an image-side surface **1082** being concave in a paraxial region thereof. The eighth lens element **1080** is made of plastic material and has the object-side surface **1081** and the image-side surface **1082** being both aspheric. The image-side surface **1082** of the eighth lens element **1080** has at least one inflection point.

(256) The IR-cut filter **1090** is made of glass and located between the eighth lens element **1080** and the image surface **1095**, and will not affect the focal length of the photographing optical lens assembly.

(257) The detailed optical data of the 10th embodiment are shown in Table 19 and the aspheric surface data are shown in Table 20 below.

(258) TABLE-US-00027 TABLE 19 10th Embodiment $f = 6.14$ mm, $Fno = 1.85$, $HFOV = 35.0$ deg. Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity 1 Lens 1 2.979 (ASP) 0.980 Plastic 1.544 55.9 4.97 2 -26.178 (ASP) 0.035 3 Ape. Stop Plano 0.035 4 Lens 2 9.046 (ASP) 0.505 Plastic 1.639 23.5 -10.00 5 3.663 (ASP) 0.404 6 Lens 3 -40.783 (ASP) 0.220 Plastic 1.639 23.5 -36.78 7 55.598 (ASP) 0.087 8 Lens 4 18.951 (ASP) 0.704 Plastic 1.544 55.9 24.01 9 -41.522 (ASP) 0.170 10 Lens 5 8.111 (ASP) 0.312 Plastic 1.544 55.9 43.60 11 12.158 (ASP) 0.268 12 Lens 6 -5.241 (ASP) 0.608 Plastic 1.544 55.9 12.02 13 -3.028 (ASP) 0.295 14 Lens 7 8.050 (ASP) 0.753 Plastic 1.544 55.9 -21.02 15 4.569 (ASP) 0.553 16 Lens 8 5.287 (ASP) 0.731 Plastic 1.544 55.9 -8.84 17 2.395 (ASP) 0.600 18 IR-cut filter Plano 0.328 Glass 1.517 64.2 — 19 Plano 0.236 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).

(259) TABLE-US-00028 TABLE 20 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k =$
1.1148E-01 -6.2724E+00 -9.0000E+01 -2.7858E+01 9.0000E+01 5.7785E+01 $A4 =$
8.8488E-04 -2.8771E-02 -4.1644E-02 2.3974E-02 -8.9616E-03 -1.1613E-03 $A6 =$
-2.6258E-03 3.7152E-02 4.3451E-02 -2.7845E-02 -3.2500E-03 2.5336E-03 $A8 =$
2.3856E-03 -2.1633E-02 -2.3216E-02 2.1016E-02 1.5573E-03 — $A10 = -1.2197E-03$
6.9571E-03 7.0528E-03 -1.0072E-02 -1.0724E-03 — $A12 = 3.0873E-04 -1.1920E-03$
-1.1181E-03 2.4771E-03 9.6120E-05 — $A14 = -3.4098E-05 8.2403E-05 7.8527E-05$
-2.5471E-04 -1.2877E-05 — Surface # 8 9 10 11 12 13 $k = -9.0000E+01 9.0000E+01$
-9.0000E+01 2.8087E+01 2.7305E+00 -1.2349E+01 $A4 = -7.2005E-03 -1.5313E-02$
-1.1144E-02 -3.6581E-02 -1.2640E-02 -4.6138E-02 $A6 = 2.0708E-03 -2.6600E-03$
-6.0561E-03 1.0973E-02 2.5183E-02 2.8088E-02 $A8 = -1.5811E-04 -1.2628E-03$
-4.9662E-03 -1.9182E-02 -1.3570E-02 $A10 = -1.2490E-04 3.1810E-04 -2.2111E-03$

7.3637E-03 3.9669E-03 A12 = — 1.5688E-05 3.7125E-05 2.3378E-03 -1.2939E-03
-6.0252E-04 A14 = — — — -6.3497E-04 7.4233E-05 3.7516E-05 A16 = — — —
5.7425E-05 — — Surface # 14 15 16 17 k = -8.6451E+00 -3.3055E+01 -4.2717E-01
-1.7325E+00 A4 = -1.8503E-02 -7.1996E-03 -9.6192E-02 -7.1224E-02 A6 = 5.1280E-03
2.1049E-03 2.0466E-02 1.7893E-02 A8 = -3.8425E-03 -1.1926E-03 -2.6027E-03
-3.3474E-03 A10 = 9.0824E-04 2.0077E-04 2.4728E-04 4.1579E-04 A12 = -1.3407E-04
-1.2997E-05 -1.7061E-05 -3.1360E-05 A14 = 9.4084E-06 2.6626E-07 7.1256E-07
1.2815E-06 A16 = — — -1.3116E-08 -2.1579E-08

(260) In the 10th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment, so an explanation in this regard will not be provided again.

(261) Moreover, these parameters can be calculated from Table 19 and Table 20 as the following values and satisfy the following conditions:

(262) TABLE-US-00029 10th Embodiment f [mm] 6.14 ($\Sigma AT + BL$)/ ΣCT 0.63 Fno 1.85 TL/ImgH 1.74 HFOV [deg.] 35.0 TL [mm] 7.82 Nmax 1.639 Yc7/f(object-side surface 0.20 271) (V2 + V6)/V1 1.42 Yc7/f(image-side surface 0.27 272) (R11 - R12)/(R11 + R12) 0.27 Yc82/f 0.28 (R15 - R16)/(R15 + R16) 0.38 f/fG1 0.77 f/R16 2.56 f/fG2 0.23 f/f1 1.23 f/fG3 -0.37 f1/f2 -0.50 f/f7 -0.29 Y11/Y82 0.51 f/f8 -0.69 SD/TD 0.85 CT3/CT4 0.31 ATmax/ImgH 0.12 CT7/CT8 1.03 f/EPD 1.85 T23/T56 1.51 BL/EPD 0.35 T23/T78 0.73 TL/EPD 2.36

11th Embodiment

(263) FIG. 21 is a schematic view of an image capturing unit according to the 11th embodiment of the present disclosure. FIG. 22 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 11th embodiment. In FIG. 21, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 1197. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 1100, a first lens element 1110, a second lens element 1120, a third lens element 1130, a fourth lens element 1140, a fifth lens element 1150, a sixth lens element 1160, a seventh lens element 1170, an eighth lens element 1180, an IR-cut filter 1190 and an image surface 1195. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 1110 and the second lens element 1120. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 1130, fourth lens element 1140 and the fifth lens element 1150. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 1160, the seventh lens element 1170 and the eighth lens element 1180. The image sensor 1197 is disposed on or near the image surface 1195 of the photographing optical lens assembly, and the photographing optical lens assembly has a total of eight lens elements (1110-1180). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has negative refractive power, and the third lens group has negative refractive power.

(264) The first lens element 1110 with positive refractive power has an object-side surface 1111 being convex in a paraxial region thereof and an image-side surface 1112 being convex in a paraxial region thereof. The first lens element 1110 is made of plastic material and has the object-side surface 1111 and the image-side surface 1112 being both aspheric.

(265) The second lens element 1120 with negative refractive power has an object-side surface 1121 being concave in a paraxial region thereof and an image-side surface 1122 being convex in a paraxial region thereof. The second lens element 1120 is made of plastic material and has the object-side surface 1121 and the image-side surface 1122 being both aspheric.

(266) The third lens element **1130** with negative refractive power has an object-side surface **1131** being convex in a paraxial region thereof and an image-side surface **1132** being concave in a paraxial region thereof. The third lens element **1130** is made of plastic material and has the object-side surface **1131** and the image-side surface **1132** being both aspheric.

(267) The fourth lens element **1140** with negative refractive power has an object-side surface **1141** being convex in a paraxial region thereof and an image-side surface **1142** being concave in a paraxial region thereof. The fourth lens element **1140** is made of plastic material and has the object-side surface **1141** and the image-side surface **1142** being both aspheric.

(268) The fifth lens element **1150** with positive refractive power has an object-side surface **1151** being convex in a paraxial region thereof and an image-side surface **1152** being convex in a paraxial region thereof. The fifth lens element **1150** is made of plastic material and has the object-side surface **1151** and the image-side surface **1152** being both aspheric.

(269) The sixth lens element **1160** with negative refractive power has an object-side surface **1161** being concave in a paraxial region thereof and an image-side surface **1162** being convex in a paraxial region thereof. The sixth lens element **1160** is made of plastic material and has the object-side surface **1161** and the image-side surface **1162** being both aspheric.

(270) The seventh lens element **1170** with positive refractive power has an object-side surface **1171** being convex in a paraxial region thereof and an image-side surface **1172** being concave in a paraxial region thereof. The seventh lens element **1170** is made of plastic material and has the object-side surface **1171** and the image-side surface **1172** being both aspheric.

(271) The eighth lens element **1180** with negative refractive power has an object-side surface **1181** being convex in a paraxial region thereof and an image-side surface **1182** being concave in a paraxial region thereof. The eighth lens element **1180** is made of plastic material and has the object-side surface **1181** and the image-side surface **1182** being both aspheric. The image-side surface **1182** of the eighth lens element **1180** has at least one inflection point.

(272) The IR-cut filter **1190** is made of glass and located between the eighth lens element **1180** and the image surface **1195**, and will not affect the focal length of the photographing optical lens assembly.

(273) The detailed optical data of the 11th embodiment are shown in Table 21 and the aspheric surface data are shown in Table 22 below.

(274) TABLE-US-00030 TABLE 21 11th Embodiment $f = 8.50$ mm, $Fno = 1.65$, $HFOV = 34.4$ deg. Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity 1 Ape. Stop Plano -0.941 2 Lens 1 3.940 (ASP) 1.372 Plastic 1.544 55.9 7.09 3 -159.104 (ASP) 0.085 4 Lens 2 -30.053 (ASP) 0.350 Plastic 1.639 23.5 -72.38 5 -86.207 (ASP) 0.035 6 Lens 3 4.758 (ASP) 0.454 Plastic 1.632 23.4 -18.99 7 3.282 (ASP) 0.650 8 Lens 4 19.021 (ASP) 0.403 Plastic 1.640 23.3 -178.90 9 16.176 (ASP) 0.286 10 Lens 5 25.289 (ASP) 1.679 Plastic 1.544 55.9 25.12 11 -29.045 (ASP) 0.407 12 Lens 6 -4.945 (ASP) 0.488 Plastic 1.639 23.5 -21.78 13 -7.966 (ASP) 0.051 14 Lens 7 4.920 (ASP) 1.346 Plastic 1.544 55.9 18.40 15 8.744 (ASP) 0.600 16 Lens 8 5.758 (ASP) 1.305 Plastic 1.544 55.9 -16.14 17 3.200 (ASP) 0.850 18 IR-cut filter Plano 0.300 Glass 1.517 64.2 — 19 Plano 0.306 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).

(275) TABLE-US-00031 TABLE 22 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k =$
 $1.2865E-01$ $5.0000E+01$ $-9.0000E+01$ $9.0000E+01$ $-2.2504E+01$ $-1.4212E+01$ $A4 =$
 $-3.4761E-04$ $-1.2266E-02$ — — $-1.4840E-02$ $1.0221E-02$ $A6 = -2.5199E-04$ $6.5425E-03$
— — $7.3482E-03$ $-4.2282E-03$ $A8 = 1.7461E-04$ $-1.7990E-03$ — — $-1.8865E-03$
 $1.7683E-03$ $A10 = -4.9841E-05$ $2.8240E-04$ — — $2.8085E-04$ $-4.1315E-04$ $A12 =$
 $6.7992E-06$ $-2.3274E-05$ — — $-2.2536E-05$ $4.7678E-05$ $A14 = -3.6949E-07$ $7.5019E-07$
— — $7.7267E-07$ $-2.0558E-06$ Surface # 8 9 10 11 12 13 $k = 2.6616E+01$ $2.9627E+01$
 $-2.8713E+01$ $6.7529E+01$ $7.4491E-01$ $-5.4044E+00$ $A4 = -9.8752E-04$ $7.2552E-04$
 $-3.2784E-03$ $-1.3704E-02$ $-4.2131E-03$ $-1.4501E-02$ $A6 = -1.7469E-04$ $-3.1698E-04$

-2.9428E-04 2.3327E-03 4.2125E-03 4.8408E-03 A8 = 8.2742E-05 -1.1546E-05
-1.5126E-05 -3.9459E-04 -1.5529E-03 -1.1139E-03 A10 = -3.9896E-05 -1.0942E-06
7.0476E-06 -8.9049E-05 2.9528E-04 1.5855E-04 A12 = 3.1285E-06 9.1424E-07
7.1799E-07 4.5113E-05 -2.5309E-05 -1.1892E-05 A14 = -6.7955E-08 — — -6.0673E-06
6.9653E-07 3.5594E-07 A16 = — — — 2.6922E-07 — — Surface # 14 15 16 17 k =
-1.4434E+01 -3.6876E+01 -6.6676E-01 -9.6814E-01 A4 = -4.8537E-03 -1.5739E-03
-3.3474E-02 -2.6279E-02 A6 = 8.0475E-04 3.0754E-04 3.4348E-03 3.0112E-03 A8 =
-3.0810E-04 -1.0049E-04 -2.1322E-04 -2.7412E-04 A10 = 3.6146E-05 8.1661E-06
9.9226E-06 1.6683E-05 A12 = -2.6803E-06 -2.5306E-07 -3.3485E-07 -6.1600E-07 A14 =
9.5162E-08 2.5346E-09 6.8565E-09 1.2309E-08 A16 = — — -6.1344E-11 -1.0117E-10

(276) In the 11th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 11th embodiment, so an explanation in this regard will not be provided again.

(277) Moreover, these parameters can be calculated from Table 21 and Table 22 as the following values and satisfy the following conditions:

(278) TABLE-US-00032 11th Embodiment f [mm] 8.50 (Σ AT + BL)/ Σ CT 0.48 Fno 1.65 TL/ImgH 1.83 HFOV [deg.] 34.4 TL [mm] 10.97 Nmax 1.640 Yc7/f(object-side surface 0.25 271) (V2 + V6)/V1 0.84 Yc7/f(image-side surface 0.28 272) (R11 - R12)/(R11 + R12) -0.23 Yc82/f 0.32 (R15 - R16)/(R15 + R16) 0.29 f/fG1 1.10 f/R16 2.66 f/fG2 -0.14 f/f1 1.20 f/fG3 -0.43 f1/f2 -0.10 f/f7 0.46 Y11/Y82 0.48 f/f8 -0.53 SD/TD 0.90 CT3/CT4 1.13 ATmax/ImgH 0.11 CT7/CT8 1.03 f/EPD 1.65 T23/T56 0.09 BL/EPD 0.28 T23/T78 0.06 TL/EPD 2.13

12th Embodiment

(279) FIG. 23 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure. FIG. 24 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment. In FIG. 23, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 1297. The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element 1210, an aperture stop 1200, a second lens element 1220, a third lens element 1230, a fourth lens element 1240, a fifth lens element 1250, a sixth lens element 1260, a seventh lens element 1270, an eighth lens element 1280, an IR-cut filter 1290 and an image surface 1295. A first lens group (reference numeral is omitted) of the photographing optical lens assembly includes the first lens element 1210 and the second lens element 1220. A second lens group (reference numeral is omitted) of the photographing optical lens assembly includes the third lens element 1230, fourth lens element 1240 and the fifth lens element 1250. A third lens group (reference numeral is omitted) of the photographing optical lens assembly includes the sixth lens element 1260, the seventh lens element 1270 and the eighth lens element 1280. The image sensor 1297 is disposed on or near the image surface 1295 of the photographing optical lens assembly and the photographing optical lens assembly has a total of eight lens elements (1210-1280). There is an air gap in a paraxial region between every two lens elements of the third lens group that are adjacent to each other. In this embodiment, the first lens group has positive refractive power, the second lens group has positive refractive power, and the third lens group has negative refractive power.

(280) The first lens element 1210 with positive refractive power has an object-side surface 1211 being convex in a paraxial region thereof and an image-side surface 1212 being concave in a paraxial region thereof. The first lens element 1210 is made of plastic material and has the object-side surface 1211 and the image-side surface 1212 being both aspheric.

(281) The second lens element 1220 with negative refractive power has an object-side surface 1221 being convex in a paraxial region thereof and an image-side surface 1222 being concave in a

paraxial region thereof. The second lens element **1220** is made of plastic material and has the object-side surface **1221** and the image-side surface **1222** being both aspheric.

(282) The third lens element **1230** with positive refractive power has an object-side surface **1231** being convex in a paraxial region thereof and an image-side surface **1232** being concave in a paraxial region thereof. The third lens element **1230** is made of plastic material and has the object-side surface **1231** and the image-side surface **1232** being both aspheric.

(283) The fourth lens element **1240** with positive refractive power has an object-side surface **1241** being convex in a paraxial region thereof and an image-side surface **1242** being convex in a paraxial region thereof. The fourth lens element **1240** is made of plastic material and has the object-side surface **1241** and the image-side surface **1242** being both aspheric.

(284) The fifth lens element **1250** with positive refractive power has an object-side surface **1251** being concave in a paraxial region thereof and an image-side surface **1252** being convex in a paraxial region thereof. The fifth lens element **1250** is made of plastic material and has the object-side surface **1251** and the image-side surface **1252** being both aspheric.

(285) The sixth lens element **1260** with negative refractive power has an object-side surface **1261** being concave in a paraxial region thereof and an image-side surface **1262** being concave in a paraxial region thereof. The sixth lens element **1260** is made of plastic material and has the object-side surface **1261** and the image-side surface **1262** being both aspheric.

(286) The seventh lens element **1270** with positive refractive power has an object-side surface **1271** being convex in a paraxial region thereof and an image-side surface **1272** being convex in a paraxial region thereof. The seventh lens element **1270** is made of plastic material and has the object-side surface **1271** and the image-side surface **1272** being both aspheric.

(287) The eighth lens element **1280** with negative refractive power has an object-side surface **1281** being concave in a paraxial region thereof and an image-side surface **1282** being concave in a paraxial region thereof. The eighth lens element **1280** is made of plastic material and has the object-side surface **1281** and the image-side surface **1282** being both aspheric. The image-side surface **1282** of the eighth lens element **1280** has at least one inflection point.

(288) The IR-cut filter **1290** is made of glass and located between the eighth lens element **1280** and the image surface **1295**, and will not affect the focal length of the photographing optical lens assembly.

(289) The detailed optical data of the 12th embodiment are shown in Table 23 and the aspheric surface data are shown in Table 24 below.

(290) TABLE-US-00033 TABLE 23 12th Embodiment $f = 8.58$ mm, $Fno = 2.50$, $HFOV = 33.1$ deg. Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity 1 Lens 1 4.025 (ASP) 1.071 Plastic 1.535 55.7 8.15 2 47.762 (ASP) 0.323 3 Ape. Stop Plano -0.070 4 Lens 2 8.074 (ASP) 0.415 Plastic 1.660 20.4 -18.33 5 4.743 (ASP) 1.269 6 Lens 3 45.587 (ASP) 0.291 Plastic 1.660 20.4 390.73 7 55.237 (ASP) 0.112 8 Lens 4 74.994 (ASP) 1.284 Plastic 1.530 55.8 75.71 9 -85.777 (ASP) 0.502 10 Lens 5 -62.668 (ASP) 1.014 Plastic 1.535 55.7 90.95 11 -27.546 (ASP) 0.077 12 Lens 6 -13.025 (ASP) 0.364 Plastic 1.583 30.2 -11.17 13 13.170 (ASP) 0.128 14 Lens 7 4.412 (ASP) 1.661 Plastic 1.530 55.8 3.83 15 -3.275 (ASP) 0.828 16 Lens 8 -9.566 (ASP) 0.466 Plastic 1.535 55.7 -3.63 17 2.475 (ASP) 1.050 18 IR-cut filter Plano 0.450 Glass 1.517 64.2 — 19 Plano 0.336 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).

(291) TABLE-US-00034 TABLE 24 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k =$
2.8146E-01 -4.9781E+01 -3.3149E+01 -1.6470E+01 -1.6022E+00 -9.0000E+01 $A4 =$
7.8489E-04 -5.1830E-03 -1.2582E-02 4.4560E-03 -3.5358E-03 -1.5971E-03 $A6 =$
-1.3362E-04 2.5389E-03 6.0690E-03 -4.8323E-04 -3.8943E-04 -6.3146E-04 $A8 =$
1.3044E-04 -5.8842E-04 -2.1962E-03 -2.3034E-04 -6.1707E-05 -4.4227E-05 $A10 =$
-3.4878E-05 7.1879E-05 6.7645E-04 7.6887E-04 -1.0760E-05 -7.4737E-06 $A12 =$
3.9427E-06 4.0742E-06 -1.1183E-04 -3.4174E-04 -1.1789E-06 2.6590E-06 $A14 =$

9.1659E-08 -1.2069E-06 6.2396E-06 5.0002E-05 7.2359E-07 Surface # 8 9 10 11 12 13
k = 9.0000E+01 4.5783E+01 7.5528E+01 6.0649E+01 9.8813E+00 -6.7498E+01 A4 =
-4.2452E-03 -9.6445E-03 -1.1147E-02 -1.4840E-02 -4.5780E-03 -1.0623E-02 A6 =
-1.5013E-04 -4.8947E-04 -7.9172E-04 8.4013E-04 2.2166E-03 2.4305E-03 A8 =
-2.3635E-05 4.8002E-05 -5.6164E-05 -1.6690E-04 -7.1648E-04 -4.9933E-04 A10 =
1.6089E-05 -1.0433E-06 1.1729E-05 -2.9549E-05 1.0484E-04 5.7349E-05 A12 =
-1.4472E-06 -1.7797E-07 2.0984E-07 1.3009E-05 -7.0625E-06 -3.3647E-06 A14 = — —
— -1.3914E-06 1.6277E-07 7.9013E-08 A16 = — — — 4.9385E-08 — — Surface # 14 15
16 17 k = -1.0764E+01 -6.0035E+00 -1.4276E+01 -6.3665E+00 A4 = -1.3272E-03
5.5529E-03 -1.9481E-02 -1.1993E-02 A6 = 6.1595E-04 -4.0632E-05 1.9029E-03
1.5301E-03 A8 = -1.5395E-04 -4.4555E-05 -9.7463E-05 -1.2139E-04 A10 = 1.4466E-05
2.9303E-06 3.5196E-06 5.9687E-06 A12 = -7.1232E-07 -7.3162E-08 -9.5955E-08
-1.7585E-07 A14 = 1.4375E-08 4.8414E-10 1.6256E-09 2.7743E-09 A16 = — —
-1.1112E-11 -1.7713E-11

(292) In the 12th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 12th embodiment, so an explanation in this regard will not be provided again.

(293) Moreover, these parameters can be calculated from Table 23 and Table 24 as the following values and satisfy the following conditions:

(294) TABLE-US-00035 12th Embodiment f [mm] 8.58 ($\Sigma AT + BL$)/ ΣCT 0.76 Fno 2.50 TL/ImgH 1.96 HFOV [deg.] 33.1 TL [mm] 11.57 Nmax 1.660 Yc7/f(object-side surface 0.35 271) (V2 + V6)/V1 0.91 Yc7/f(image-side surface — 272) (R11 - R12)/(R11 + R12) -180.78 Yc82/f 0.42 (R15 - R16)/(R15 + R16) 1.70 f/fG1 0.68 f/R16 3.47 f/fG2 0.23 f/f1 1.05 f/fG3 -0.19 f1/f2 -0.44 f/f7 2.24 Y11/Y82 0.42 f/f8 -2.37 SD/TD 0.86 CT3/CT4 0.23 ATmax/ImgH 0.22 CT7/CT8 3.56 f/EPD 2.50 T23/T56 16.48 BL/EPD 0.53 T23/T78 1.53 TL/EPD 3.37

(295) The foregoing image capturing unit is able to be installed in, but not limited to, an electronic device, including smart phones, tablet personal computers and wearable devices.

(296) According to the present disclosure, the photographing optical lens assembly includes, in order from the object side to the image side, the first lens group, the second lens group and the third lens group. The first lens group includes two lens elements, the second lens group includes three lens elements, and the third lens group includes three lens elements. Therefore, the refractive power distribution of the first lens group, the second lens group and the third lens group is favorable for adjusting the optical parameters in a design of the photographing optical lens assembly so as to effectively improve the image quality. When specific condition is satisfied, it is favorable for reducing the back focal length of the photographing optical lens assembly, and thereby maintaining a compact size thereof. The optical parameters of the photographing optical lens assembly with some features of the lens elements, such as curvature radii of the lens elements, are favorable for applying the photographing optical lens assembly to both the conventional electronic device and the high-end electronic device. According to the present disclosure, the photographing optical lens assembly simultaneously satisfies the requirements of good image quality and compact size even though it is applied to the electronic device with high-end specifications.

(297) The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-24 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are

not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

Claims

1. A photographing optical lens assembly comprising eight lens elements, the eight lens elements being, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element, and each of the eight lens elements having an object-side surface facing toward the object side and an image-side surface facing toward the image side; wherein a total number of lens elements in the photographing optical lens assembly is eight, the first lens element has positive refractive power, at least one of the object-side surface and the image-side surface of the sixth lens element is aspheric, the image-side surface of the eighth lens element is concave in a paraxial region thereof, the image-side surface of the eighth lens element is aspheric and has at least one inflection point, and there is an air gap in a paraxial region between each of all adjacent lens elements of the photographing optical lens assembly; wherein a focal length of the photographing optical lens assembly is f , an entrance pupil diameter of the photographing optical lens assembly is EPD, a maximum value among refractive indexes of all lens elements of the photographing optical lens assembly is $N_{\max}$, and the following conditions are satisfied: $f/EPD < 2.6$; and $1.55 < N_{\max} < 1.7$.
2. The photographing optical lens assembly of claim 1, wherein the object-side surface of the second lens element is convex in a paraxial region thereof, the image-side surface of the second lens element is concave in a paraxial region thereof, the object-side surface and the image-side surface of the seventh lens element are both aspheric, and the object-side surface of the eighth lens element is aspheric.
3. The photographing optical lens assembly of claim 1, wherein the object-side surface of the eighth lens element is convex in a paraxial region thereof.
4. The photographing optical lens assembly of claim 1, wherein the seventh lens element has positive refractive power, and the image-side surface of the seventh lens element is convex in a paraxial region thereof.
5. The photographing optical lens assembly of claim 1, wherein the eighth lens element has negative refractive power; an axial distance between the object-side surface of the first lens element and an image surface is TL , and the following condition is satisfied: $TL < 12.0\text{ millimeters}$.
6. The photographing optical lens assembly of claim 1, wherein the sixth lens element has negative refractive power, the seventh lens element is made of plastic material, and the eighth lens element is made of plastic material.
7. The photographing optical lens assembly of claim 1, wherein half of a maximal field of view of the photographing optical lens assembly is $HFOV$, and the following condition is satisfied: $30.^{\circ} < HFOV < 50.^{\circ}$.
8. The photographing optical lens assembly of claim 1, wherein a curvature radius of the image-side surface of the eighth lens element is R_{16} , and the following condition is satisfied: $1.0 < f/R_{16} < 4.5$.
9. The photographing optical lens assembly of claim 1, wherein a focal length of the seventh lens element is f_7 , and the following condition is satisfied: $0.74 \leq f/f_7 \leq 1.78$.
10. The photographing optical lens assembly of claim 1, wherein a central thickness of the third lens element is CT_3 , a central thickness of the fourth lens element is CT_4 , and the following condition is satisfied: $0.23 \leq CT_3/CT_4 \leq 1.35$.
11. The photographing optical lens assembly of claim 1, wherein an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the fifth lens element and the sixth lens element is T_{56} , and the following condition is satisfied:

$1.46 \leq T_{23} / T_{56} \leq 4.93$.

12. The photographing optical lens assembly of claim 1, wherein a sum of axial distances between each of all adjacent lens elements of the photographing optical lens assembly is ΣAT , an axial distance between the image-side surface of the eighth lens element and an image surface is BL , a sum of central thicknesses of all lens elements of the photographing optical lens assembly is ΣCT , and the following condition is satisfied: $(\text{.Math. } AT + BL) / \text{.Math. } CT < 0.8$.

13. The photographing optical lens assembly of claim 1, wherein a curvature radius of the object-side surface of the eighth lens element is $R15$, a curvature radius of the image-side surface of the eighth lens element is $R16$, and the following conditions are satisfied: $f / EPD \leq 1.95$; and $-0.5 < (R15 - R16) / (R15 + R16) < 2.5$.

14. The photographing optical lens assembly of claim 1, wherein an Abbe number of the first lens element is $V1$, an Abbe number of the second lens element is $V2$, an Abbe number of the sixth lens element is $V6$, and the following condition is satisfied: $(V2 + V6) / V1 < 1.0$.

15. The photographing optical lens assembly of claim 1, wherein at least one of the object-side surface and the image-side surface of the seventh lens element has at least one non-axial critical point; a vertical distance between the at least one non-axial critical point and an optical axis is $Yc7$, and the following condition is satisfied: $0.10 < Yc7 / f < 0.60$.
